DATA MINING METHODOLOGY AND ITS APPLICATION TO LIFE INSURANCE

By

OPEYEMI OMOLOLU OKESOLA

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Declaration

I, OPEYEMI OMOLOLU OKESOLA, do hereby declare that this project is entirely my work and my composition. The work embodied in this project has not been submitted in candidature for any degree and is not concurrently being submitted for any other degree. All references made to work of other persons have been duly acknowledged.

_____

Opeyemi Okesola
August 2021

Certification

I certify that this research work was carried out by OPEYEMI OMOLOLU OKESOLA in the Department of Computer Science, College of Postgraduate Studies (COPOS), Caleb University, Lagos. The research work is considered adequate in partial fulfilment of the requirement for the award of MSc in Computer Science.

---------- ----------

DR. AYORINDE ODUROYE **DATE**
Project Supervisor

---------- ----------

DR. OLUMOYE MOSUD **DATE**
Head of Department

---------- ----------

PROF. TEJU SOMORIN **DATE**
Dean COPOS

---------- ----------

External Examiner **DATE**

Dedication

I dedicate this project to God Almighty the giver of life and all wisdom; He is good unto me, and I cannot deny it and to the memory of my beloved father Biodun Oladoja Okesola who left us nineteen years ago without saying goodbye. Daddy the seed you planted is budding fruit, the size and sweetness are awaiting the moments of time.

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Abstract

Massive amount of data is generated daily and there is a growing need to transform them into useful information and knowledge for a wide range of application in commercial and business environments. Data mining is the knowledge discovery process of discovering correlations, patterns, and trends in large volumes of data from various perspectives and summarizing it into useful information. Data mining is established discipline within the domain of Artificial Intelligence (AI) and Knowledge Engineering (KE) with its roots in algorithms, machine learning, pattern recognition and statistics but encompasses other areas of computer science. The advancement in computing and networking resources like parallel, distributed and grid technologies have made data mining possible.

The insurance industry has been growing and collect immense data from her customer's daily. Little insight is available on these data. For any insurance company to be profitable there is need to have structure to fundamentally conduct proper risk assessment, eliminate fraud and have analytical CRM in place. It is on these structure that strong investment, proper management, and governance can be built or sustained.

The purpose of this study is to improve the efficiency of life insurer through the application of data mining. To achieve this, we applied engineering design process to develop a methodology that would ensure data mining is applied

across the three key arears of risk assessment, fraud elimination, analytical CRM and how this can be established as a practice.

The proposed methodology describes the merit and demerit of applying data mining in insurance, it is vendor-neutral and put the responsibility of tools and techniques to be used on the miner based on budget and experience. The study provides recommendations for future research in the application of data mining to facilitate decisions relevant to life insurance.

Keywords

Knowledge discovery, data mining, customer relationship management, data warehouse, data analytics, decision making, life insurance, risk prediction, fraud detection

CHAPTER 1

INTRODUCTION

1.0 Introduction to the Research

We live in a world where massive volumes of data are constantly being generated and collected daily, they pour into the computer network. Every day, people use the World Wide Web (WWW) and various data storage devices in social media, commerce, society, science and engineering, medical, and other fields in every other aspect of daily life. We can say that we are in the data age. This explosion in the data is a product of the computerization of the society and the rapid development of robust data collection and storage tools.

In the modern age, virtually all automated systems generate some sort of data either for diagnostic or analysis purposes. This has resulted during a deluge of knowledge, which has been reaching the order of petabytes or exabytes. Communities and social media have become increasingly important data sources, producing digital pictures and videos, blogs, tweets, social-network interactions, and various kinds of social networks activities. The number of documents on the indexed Web is in trillions while the deep web is much larger. User accesses to such documents create access logs at servers and customer behavior profiles at commercial sites. The documents and links online can be mined to determine associations between different topics on the web

while user access logs can be mined to determine frequent patterns of accesses or unusual patterns of unwarranted behavior.

Businesses globally generate gigantic data sets, including sales transactions, stock trading records, product descriptions, sales promotions, company profiles and performance, customer feedback and using the automated teller machine (ATM) or a credit card, can create data which can be mined for many useful insights such as fraud or other unusual activity.

Scientific and engineering practices generate high orders of petabytes of data in a continuous manner, from remote sensing and the Internet of Things (IoT), process measuring, scientific experiments, system performance, engineering observations, environment surveillance and development of low-cost wearable sensors, smartphones, and other smart devices that can communicate with one another generates low-cost wearable sensors, smartphones, and other smart devices that can communicate with one another.

Telecommunication networks has quickly evolved from offering local and long-distance telephone services to providing many other comprehensive communication services such as internet access, email, text messages, images, data transmissions, and other data traffic. The integration of telecommunication, network, internet, and various other means of communication and computing has been under way, changing the face of telecommunications and computing. This has created a great demand for DM

to help understand business dynamics, make better use of resources, and improve service quality.

Global telecommunications backbone carries tens of petabytes of data traffic every day. The medical and health industry generates tremendous amounts of knowledge from medical records, patient monitoring, and medical imaging. Billions of Web searches supported by search engines process tens of petabytes of knowledge daily. The list of sources that generate huge amounts of knowledge is endless. 1.7 MB of knowledge is made every second by every one during 2020, 2.5 quintillion bytes of knowledge are produced by humans a day , every day, 306.4 billion emails are sent, and 500 million Tweets are made Bulao, J., (2021).

This explosively growing, widely available, and gigantic body of knowledge makes our time truly the info age. Powerful and versatile tools are badly needed to automatically uncover valuable information from the tremendous amounts of knowledge and to rework such data into organized knowledge.

This rapid climb of knowledge has brought new challenges to current data management systems — the relational data model — and has emphasized the necessity for a paradigm shift in technology design and development. A greater challenge is ability to process and analyse these data, this necessity has led to the birth of DM. DM has and will continue to make great strides in our

journey from the data age toward the coming information age. The ability to form data-driven decisions is crucial to any business.

The emerging field of DM is fast becoming the most progressive and promising field for data extraction and manipulation to generate useful information. Many companies use DM applications to manipulate, identify and extract useful information from records stored in their databases, data repositories, and data warehouses to uncover previously unknown patterns and relationships.

The digital revolution has allowed the collection and storage of large and diverse amounts of information referred to as big data, which is too complex for traditional data processing to be handled. These data come as either unstructured and/or structured formats and influence life insurance processing, the ability to improve the business process, the development of cross-selling and marketing strategies, the exposure of fraud, the discovery of patterns in users' access to websites and the application of patterns, relationships and trends that have been hidden or undiscovered within colossal amounts of data.

Many industries have yet to fully exploit the power of DM. Insurance, for example, is one of the new fields in which DM tools and techniques are starting to be used successfully. Underwriting, rating, pricing, understanding the interests of customers, designing risk-based insurance premiums, determining

likely churn customers on the basis of their characteristics preventing and detecting fraud in insurance claims management and dematerialization are areas where DM tools such as machine learning, neural networks, genetic algorithms, decision trees, and data visualization can be used.

However, the implementation of DM techniques is in practice inconsistent. Because software vendors offer different and proprietary approaches that focus on specific business applications. These approaches even use a variety of analysis tools. In order to develop good DM strategies, the insurer needs an application-neutral methodology. Moreover, DM approaches too often fail to keep the objectives of the organization in mind, so that the results of the DM project are irrelevant. Moreover, the system perspective is not maintained; therefore, essential components of the organizational system are ignored and, again, the results of DM are not as effective as they should be. What is needed is a guide through a labyrinth of tools and approaches to a myriad of applications.

1.1 Background to the study

DM is often described as a process of finding correlations, patterns, trends, or relationships by searching through a large amount of data stored in repositories, corporate databases, and data warehouses. The types of relationships that exist are thought to be sometimes unclear to information analysts, because the amount of information is too large or the kind of

relationship is too difficult to imagine. In that sense, humans are limited by the overload of information; therefore, new tools and techniques are being developed to solve this problem through automation.

DM uses a range of pattern recognition technologies and statistical and mathematical techniques to discover the possible rules or relationships that govern the data in the databases. DM must also be seen as an iterative process requiring targets and targets to be specified Chapman, P., et al (2000), Coenen, F., (2011) and Derrig, A., (2002)). Once the intended objectives have been fully defined, it is necessary to determine which data is available or can be collected. Sometimes data is available in data warehouses, but some cleaning and filtering is done before it can be used to convert it into information.

DM also includes a methodology for implementation. The methodology, or structured approach, varies from vendor to vendor. For example, the SAS Institute (2020), Azevedo, A., & Santos, M., (2008) promotes SEMMA (Sample, Explore, Modify, Model and Assess). CRISP-DM (CRoss-Industry Standard Process for DM) by the consortium Daimler Chryrler, SPSS and NCR, Inc Azevedo, A., & Santos, M., (2008), Bulao, J., (2021), Hearst, M., (1999)) is another methodology. Each methodology seeks to help obtain the best data to provide the most responsive information to meet their needs.

The recognition that effective decisions are based on appropriate information from accurate and up-to-date data is not new. The evolution and development

of finding the right data for decision-making began more than 30 years ago and has progressed through several stages of development.

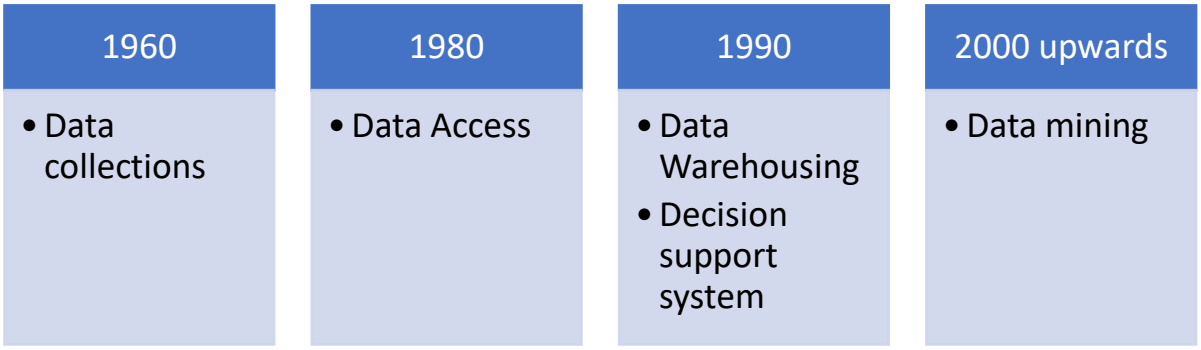


Figure 1: Data Mining Evolution

The evolutionary stages of DM are as follows:

1. Data Collection: Simple reports of pre-formatted information were made from data kept in databases from the late 1960s. The information gathered was used to make judgments based on a retrospective analysis. The data was kept in these databases, and the data was retrieved and modified by applications to create structured reports.

2. Data Access: In the 1980s, users began to want information more frequently and they wanted it to be more individualized. Thus, they began to make queries, or informational requests, of the databases. These was aided with the advent of Relational Database management System (RDBMS), Structured

Query Languages (SQL) and Open Database Connectivity (ODBC) to perform dynamic data delivery.

3. Data Queries: In the 1990s, there is a need for dynamic data delivery at multiple levels of organisation, constant and immediate access to more detailed information is required "on the fly". User wanted information to be "just-in-time" and want to drill down on details for decision-making processes. This calls for users to write their own queries and extract the information that they needed from the database.

4. Data Mining: In the last few years, users began to realize the need for more tools and techniques to identify and find relationships in data so that the information obtained was more meaningful for their applications. Advancement in multiprocessor, computer, algorithms, massive databases provide opportunities for users to adopt the needed new tools to extract information and find meaningful relationships or patterns from huge amounts of data that, if done in a traditional way, would require much time and many resources to find. The information delivery in this era is proactive and predictive in nature.

DM is a union of a number of fields and disciplines, including Database Management System (DBMS), Artificial Intelligence (AI), Machine Learning (ML). Knowledge Engineering, Statistics and Pattern Recognition.

DM trends are data and computational trends. DM in the early days works best on numerical data collected from a single database, which is grossly inadequate to work with non-numeric data, and therefore there is a need to differentiate DM from access to information Coenen, F., (2011), over the years the union of statistics, machine learning and various algorithms to mine information from non-numeric data Derrig, R., (2002), Dockrill, M., (2001). The development of fourth generation programming languages and various related computing techniques such as AI, ML and pattern recognition, computational linguistics, induction, compression, approximation and algorithms have influenced the ability to mine large volumes of heterogeneous data stored in data warehouses.

1.2 Statement of problem

Insurers collect enormous amounts of data about their clients but are unable to get and create value from the data which would made the more competitive and customer centric. There is a serious data analytical challenge in life insurance which makes it imperative for insurers to gain deep understanding of business using DM. DM methods solve data preparation challenges by automatic handling of missing values, detect nonlinearity, outliers and data errors are catered for while effectively select predictors automatically.

Life insurers underwriting has to do with the selection of subjects for life insurance in such a manner that general company objectives are met. The profitability of underwriting depends on three key area namely:

- Risk assessment: risk selection, probability of claim, expected losses on claims.
- Fraud: detecting organized professional fraud and identifying possible individual fraud.
- Analytical CRM: precision focused marketing, lapse likelihood grading, and upsell opportunities identification.

Hence the only way a insurer can be profitable is to have mechanism in place for proper risk assessment, protect the company against high claim cost due to fraudulent claims and have efficient and proactive CRM analytics in place. A bad risk selection can lead to serious problem, an increasing claim cost can dig a hole in the button line while a poor CRM analytics can result in loss of revenue and missed opportunities. Efficient and proactive supervision at an early stage could help avert these problems (Dockrill 2001). While insurers have increased their efforts these three key areas much more can still be done with the help of DM.

Although insurers exist to help society by managing risk, the way they estimate risk and the repercussions of their decisions are already being scrutinized and challenged. Companies cannot afford any form of adverse selection and are seeking for ways to offer coverage at a cost that accurately reflect its actual risk exposure and are seeking for ways to protect itself from adverse selection,

including accurately identifying risk factors, having a system for verifying information, and placing caps on coverage.

Fraudulent claims are a serious financial burden on insurers and result in higher overall life insurance costs Insurers lose millions of Naira yearly to fraudulent claims, largely because they do not having a simple means to tell which claims are valid and which are potentially false. Life insurance fraud has an impact not only on the financial health of insurers, but also on the lives of innocent people who are looking for affordable life insurance. The business of life insurance is to pay claims in a timely and efficient manner. Insurers are well aware that claimants may have opportunities and incentives to take advantage or even fabricate claims to obtain payments they might otherwise not deserve Derrig, R., (2002).

Analytical CRM is the fusion of data and intelligence. The profitability of the life insurance sector is mainly depend on the services it offers and on meeting the customer demand on a regular basis, it suggests that a the foundation of the insurance industry must be a robust CRM program. In order for a customer-centric approach to succeed, insurers must have a thorough understanding of customer behavior and responsiveness.

These three key factors if addressed by the insurers would make her profitable. To address this deficiency in life insurance, this research proposes to do the following:

- Develop a convenient methodology for the application of DM in life insurance with focus in the three key areas earlier mentioned.
- Analyze and compare the different and successfully applied tasks and techniques used in DM in the three key areas.
- Identify the main advantages and disadvantages for the application of DM techniques and tools in life insurance in the three key areas.
- Identify possible problem areas or issues for the application of DM in life insurance in the three key areas.
- Identify possible applications of DM in insurance along the three key areas.

1.3 Research Aim & Objectives

1.3.1 Research Aim

The aim of this study is to develop a DM methodology for life insurance companies to do proper risk assessment, eliminate fraud and have analytical CRM in place.

1.3.2 Research Objectives

1.3.2.1 General Research Objectives

The general objective of this study is applying the DM techniques for conducting sound risk assessment, predicting, and detecting fraudulent

insurance claims and providing an analytical CRM while ensuring the company is stable, agile and highly adaptable to the constantly changing market environment.

1.3.2.2 Specific Research Objectives

For the realization of the general objective stated above, the following specific objectives are formulated.

- To review literature on DM technology and their application in the insurance industry for the purpose of getting information that will help in this research.
- To identify sources and collect the required data from life Insurance
- Develop a convenient methodology for the application of data mining in life insurance.
- Examine and compare the various data mining tasks and strategies that have been effectively implemented.
- Identify the main advantages and disadvantages for the application of data mining techniques and tools in life insurance.
- Identify possible problem areas or issues for the application of data mining in insurance.
- Identify possible applications of data mining in the field of insurance.

- To report findings of the result and forward recommendation for further research.

1.4 Significance of the Study

DM is a key tool used in many areas of life that has made a positive contribution to all areas of human endeavor by providing insight into various unrelated data. The findings of this study will benefit African Alliance Insurance PLC and other insurers. Information is the new oil; access and correlation would provide a superior competitive edge, lower insurance costs, provide a fair price for customers, reduce time on the market, and make insurance in Nigeria desirable and profitable.

Specifically, it would:

- To save time and money which is wasted for conducting risk assessment, processing, and surveying claims and getting insight into who your current and potential customers are.
- To improve the decision-making process for experts especially in the three-key focus of this research.
- To improve rules and procedures that are related to risk selection, computing probability of claims, claims handling, targeted marketing, cross-selling, and upselling.
- To create a smooth relation with her clients

This study can also be an input for further research in this and other related areas in the context of our country. Finally, this study would give hands on experience for the researcher for understanding studies in the future.

1.5 Scope and limitations of the Study

This study focuses specifically on the application of DM to the issues of risk assessment, fraudulent claims and analytical CRM which are critical areas. It emphasizes the application of a system analysis and design perspective to develop a DM methodology that is appropriate for such applications. Only the information necessary to illustrate the concepts described in this document would have been included. For that reason, requirements necessary for the application of DM in other fields have been omitted.

The methodology proposed for this study is an abstract and functional framework. It is a conceptual model, has not yet been implemented and has therefore not been executed or tested. That the task remains to the future.

1.6 Structure of this Paper

This thesis is divided into five chapters, including the introductory chapter. Chapter 2, reviews some of the most important cases of DM projects in areas such as underwriting, rating, pricing, risk-based insurance premium design, determination of likely churn customers on the basis of their characteristics to prevent and detect fraud in claims management, dematerialization of insurance and product development. Chapter 3 provides a general description

of the research methodology used to address the issues identified earlier in this chapter. Chapter 4 sets out the methodology for the use of DM in insurance engineering. Chapter 5 summarizes the main conclusions of this document. It presents the main advantages and disadvantages for the application of DM techniques and insurance tools, as well as areas of concern or problems for their implementation. Finally, this chapter also sets out areas for further research.

1.7 Definition of Terms

Attrition: Retention of Customers

Beneficiary - The beneficiary of a insurance policy is the person or people named by the insured who receive the death benefit upon the death of the insured.

CRM (Customer Relationship Management): Process of studying and interacting with customers in order to maximize profits. It includes ensuring customer satisfaction as well as cutting service to unprofitable customers.

Data Mart: Small data warehouse focused on a single area such as a research project or department.

Data Repositories: Collection of resources that can be used to retrieve information, repositories usually consist of several databases tied together by a common search engine.

Data Visualization: Techniques for turning data into information by using visual representation and the capacity of the human brain to recognize patterns and trends.

Data Warehouse: A static copy of a database generally optimized for analysis or renormalized.

Death benefit - This is the money paid to the beneficiaries when the insured dies.

Decision Trees: Once of the most popular data mining techniques, that search for ways to divide data into subgroups that are as much similar as possible with regard to a target variable. Two of the most popular tree models are the CHAID and the CART.

Genetic Algorithms: A computer-based method of generation and testing combination of input parameters to find an optimal output. This process is based on natural evolution concepts such as combination, mutation, and natural selection.

Induction: Technique that infers generalizations from the information in data.

Insured - The individual whose life is insured by the insurance policy. Upon this person's death, a benefit is paid to the named beneficiaries if the policy is in force.

Neural Network: Model that mimic the brain through systems of equations, they learn by being trained with a data set.

Permanent insurance policy - Permanent insurance does not expire as long as required premium payments are made on time. They are designed to last your entire lifetime and often accrue cash value over time.

Premiums - Premiums are simply payments you make to keep your insurance policy in force. Your insurance policy will remain in force until the end of the contract or until you die, if required premium payments are made on a timely basis.

Term insurance - This type of insurance is temporary. It provides a given amount of coverage for a specified period of time, which ends at the expiration date set in the policy.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The convergence of modern technologies of computers, networks, and sensors have made data collection and organization an almost effortless task. The data generated abound in its raw form, these the captured data need to be converted into information and knowledge from recorded data to become useful. Data, which is growing at an unprecedented rate, is becoming an inextricable part of most people's everyday lives around the world, according to the World Bank (2021). These data include textual content (i.e. structured, semi-structured as well as unstructured), to multimedia content across a variety of platforms (videos, photos, and audio) machine-to-machine communications, social media sites, sensors networks, cyber-physical systems, and Internet of Things [IoT]) Sivarajah, U., Kamal, M., Irani, Z., Weerakkody, V., (2017) There is a huge amount of information locked up in databases—information that is potentially important but has not yet been discovered or articulated.

The traditional task of extracting useful information from recorded data has been performed by analysts; however, the increasing volume of data in modern businesses and sciences calls for computer-based methods for this task. As data sets have grown and complexity, so there had been an inevitable shift away

from direct hands-on data analysis toward indirect, automatic data analysis in which the analyst works via more complex and sophisticated tools. The entire process of applying computer-based methodology, including new techniques for knowledge discovery from data, is often called DM.

DM is a powerful tool that can help to find patterns and relationships within our data Sharma, A., Sharma, M., & Dwivedi, R., (2017). In addition, Hilage, T., and Kulkarni, R., (2012) defines DM is used for mining data from databases and finding out meaningful patterns from the database. To achieve this field many discipline and field are involved hence Umamaheswari, K., & Janakiraman, D., (2014) view DM as an interdisciplinary field of astronomy, business, and computer science, to find new trends in massive datasets, whether economic and otherwise. DM is not a simple process, hence many tools and processes are involved hence Witten, I., Frank, E., Hall, M., & Pal, C., (2017) describes DM as the extraction of implicit, previously unknown, and potentially useful information from data and further proposed that to achieve this, there is a need to have a computer programs that sift through databases automatically, seeking regularities or patterns and in the process discover strong patterns which can be used to make accurate predictions on future data. There is however no certainty that the result would be exact, there would be exception to every rule, few outliers may exist, but the algorithms must be robust enough to cope with imperfect data and to extract regularities that are inexact but

useful. To mine data, there is need to develop a number of algorithms in statistics, machine learning, and artificial intelligence domains. As such, DM is defined as a set of rules, processes, algorithms that are designed to generate actionable insights, extract patterns, and identify relationships from large datasets Morabito, V., (2016).

The definition by Berry, M., & Linoff, G., (2000) seem to appeal more to the real purposes, means, techniques and tools of DM. They define DM as the process of automatically or semiautomatically exploring and analyzing massive amounts of data to identify significant patterns and rules. Some key characteristics of DM in this definition are that large quantities of data must be involved, patterns and rules to be found ought to be known and meaningful, that there is need to focus on the exploration and analysis.

From the above, it can be deduced that DM have very wide application, requires various skillset, it is not a product to be brought rather it is a discipline that must be mastered.

2.2 Data Mining and Knowledge discovery

We live in a world that generates and collects massive volumes of data. The past century saw an exponential growth in data, there have been rapid pace of change in nearly every area, but none was more pronounced as the fast pace in the quantity of stored information. In every industry, from manufacturing to

medical to marketing, this information boom has brought new opportunities and challenges.

As the volume of data increases, the proportion of information in which people could understand decreases substantially. This reveals that the level of understanding of people about the data at hand could not keep pace with the rate of generation of data in various forms, which results in increasing information gap Adane, T., (2011). Many research have been conducted to discover this bottleneck and to look into possible remedies Plotnikova, V., Dumas, M., & Milani, F., (2020). Current technological progress permits the storage and access of large amounts of data at virtually no cost. Although it has been taught many times, the fundamental difficulty in today's information-centric world is putting the obtained raw data to good use Kurgan, A., and Musilek, P. (2006). The true value is not in storing the data, but rather in our ability to extract useful reports and to find interesting trends and correlations, through the use of statistical analysis and inference, to support decisions and policies made by scientists and businesses Fayyad, U., et al. (1996). To bridge the gap of analyzing large volume of data and extracting useful information and knowledge for decision making that the new generation of computerized methods known as Data Mining (DM) or Knowledge Discovery in Databases (KDD) has emerged in recent years.

The foundation of modern DM methodologies were first proposed by Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P., (1996a, 1996b, 1996c), and were initially related to Knowledge Discovery in Databases (KDD). KDD presents a conceptual process model of computational theories and tools that support information extraction (knowledge) with data Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P., (1996a). In KDD, the overall approach to knowledge discovery includes DM as a specific step. As a result, KDD's nine primary steps (exhibited in Figure. 2), has the advantage of considering data storage and access, algorithm scaling, interpretation and visualization of results, and human computer interaction Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P., (1996a, 1996c). Introduction of KDD also formalized clearer distinction between DM and data analytics, as for example formulated in Tsai, C., et al, (2015): "…by the data analytics, we mean the whole KDD process, while by the data analysis, we mean the part of data analytics that is aimed at finding the hidden information in the data, such as DM".

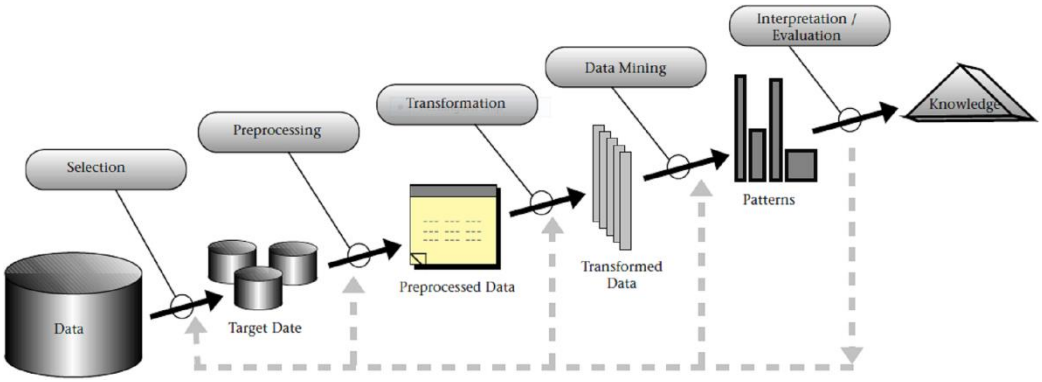


Figure 2: An overview of the steps composing the KDD process as presented in Fayyad, Piatetsky-Shapiro & Smyth (1996a, 1996c)

The main steps of KDD are as follows:

Step 1: Learning application domain: The first phase is to gain a thorough understanding of the application domain and any relevant prior knowledge, followed by determining the KDD process's purpose from the customer's perspective.

Step 2: Dataset creation: The second stage is to choose a dataset, focusing on a selection of variables or data samples on which to do discovery.

Step 3: Data cleaning and processing: In the third step, basic operations to remove noise or outliers are performed. The gathering of data needed to model or account for noise, deciding on techniques for dealing with missing data fields, and accounting for data types, schema, and mapping of missing and unknown values are all taken into account.

Step 4: Data reduction and projection: This is where the process of finding meaningful features to represent the data is done, and depending on the task's

aim, transformation methods are used to discover the best features set for the data.

Step 5: Choosing the function of data mining: The goal outcome (e.g., summarization, classification, regression, clustering) is defined in the fifth stage.

Step 6: Choosing data mining algorithm: The sixth phase entails picking which method(s) to use to look for patterns in the data, determining which models and parameters are appropriate, and connecting a specific data mining approach to the KDD process's overall requirements.

Step 7: Data mining: The job of data mining, or searching for patterns of interest in a particular representational form or a set of such representations: classification rules or trees, regression, and clustering, is carried out in the seventh phase.

Step 8: Interpretation: The redundant and unnecessary patterns are filtered out in this step, and the relevant patterns are analyzed and presented in a form that the consumers can understand.

Step 9: Using discovered knowledge: The results are then incorporated into the performance system, documented and communicated to stakeholders, and utilized as the basis for choices in the final stage.

The KDD process became dominant in industrial and academic domains Kurgan, A., and Musilek, P., (2006) and Marban, O., Mariscal. G., Segovia. J.,

(2009). The timeline of data mining methodologies evolution and process models shows in Fig 3 below.

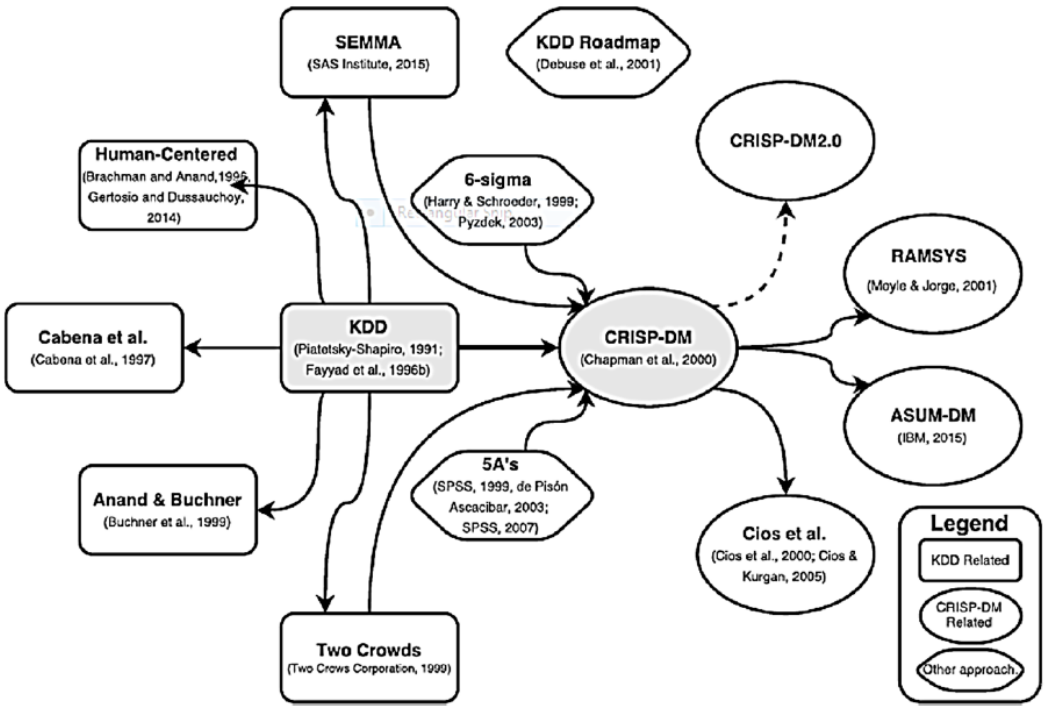


Figure 3: Evolution of data mining process and methodologies, as presented in Martnez-Plumed et al. (2017)

Other techniques and process models were constructed on the foundation of the original KDD data mining paradigm. Other models extend the KDD process to meet various gaps and weaknesses, ranging from process reform to a total shift in focus. Plotnikova, V., Dumas, M., Milani, F., (2020). Brachman, R., Anand, T., (1996) and later Gertosio, C., Dussauchoy, A., (2004) suggested workable adjustments to the process to cater for the interactivity and iterative nature of the process. They held that for the KDD process to be complete, it needs to be enhanced with supplementary task and the focus must change to a

user's point of view (human-centered approach) with focus on the decisions that need to be made by the user in the course of data mining process.

In contrast, Cabena, P., et al., (1997) proposed different number of steps emphasizing and detailing data processing and discovery tasks. Anand, S. & Büchner, A. (1998), Anand et al. (1998), Buchner, A. et al. (1999) presented additional data mining process steps by focusing on of data mining process adaptation to business settings. They focused on cross-selling to online customers, incorporating the internet data finding process as well (web-based mining). Two Crows data mining process model is consultancy originated framework that has defined the steps differently but is still close to original KDD. Finally, SAS Institute Inc. developed SEMMA (Sample, Explore, Modify, Model, and Assess) in 2005, which is based on KDD (2017). It is characterized as a logical structure of SAS Enterprise Miner's functional toolset for performing data mining's main activities. This is a vendor-specific process model, as opposed to KDD, which limits its implementation in multiple situations. SEMMA omits two steps from the original KDD approach ('Learning Application Domain' and 'Using Discovered Knowledge'), both of which are considered critical to a data mining project's success. Mariscal, G., Marbán, Ó., Fernández, C., (2010). In terms of adoption, new KDD-based proposals received limited attention across academia and industry Kurgan, A., & Musilek, P., (2006); Marban, O., Mariscal, G., Segovia, J., (2009).

Subsequently, most of these methodologies converged into the CRISP-DM methodology Plotnikova, V., Dumas, M., & Milani, F., (2020).

In response to changing needs and early market interest in data mining, an industry-driven methodology called Cross-Industry Standard Process for Data Mining (CRISP-DM) was introduced as an alternative to KDD. The consortium was funded by the European Commission to develop a standard process model that is domain-agnostic Mariscal, G,, Marbán, Ó,, Fernández, C., (2010), non-proprietary, freely available and would address the issues around the initial KDD model. These distinctive characteristics have made CRISP-DM to be considered as 'de-facto' standard of data mining methodology and as a reference framework to which other methodologies are benchmarked Mariscal, G,, Marbán, Ó,, Fernández, C., (2010). The main steps of CRIPS-DM, as depicted in Fig. 3 below are as follows:

Phase 1: Business understanding: The first step focuses on gaining a business understanding of the project objectives and needs, which is then translated into data mining problem definitions. This initial step also includes the presentation of a tentative plan to attain the goals.

Phase 2: Data understanding: This process begins with data gathering and progresses through actions to familiarize yourself with the data, find data quality issues, get initial insights into the data, and maybe detect and establish hypotheses.

Phase 3: Data preparation: The operations required to generate the final dataset from the initial raw data are covered in the third step. Data preparation tasks are repeated on a regular basis.

Phase 4: Modeling phase: Various modeling techniques are chosen and applied in this step, followed by the calibration of their parameters. For the same data mining problem, several strategies are usually utilized..

Phase 5: Evaluation of the model(s): The fifth phase starts with a quality perspective and then ensures that the model(s) meets the business objectives before moving on to final model deployment. A decision on how to use the data mining results should be made at the end of this step.

Phase 6: Deployment phase: The models are then deployed to allow end users to use the data as a basis for choices or to aid in the business process. Even though the model's goal is to improve data knowledge, the information gathered must be arranged, presented, and disseminated in a way that the end user can understand. The deployment phase might be as simple as gene editing, depending on the needs. The deployment phase might be as easy as generating a report or as sophisticated as implementing a repeatable data mining process, depending on the requirements.

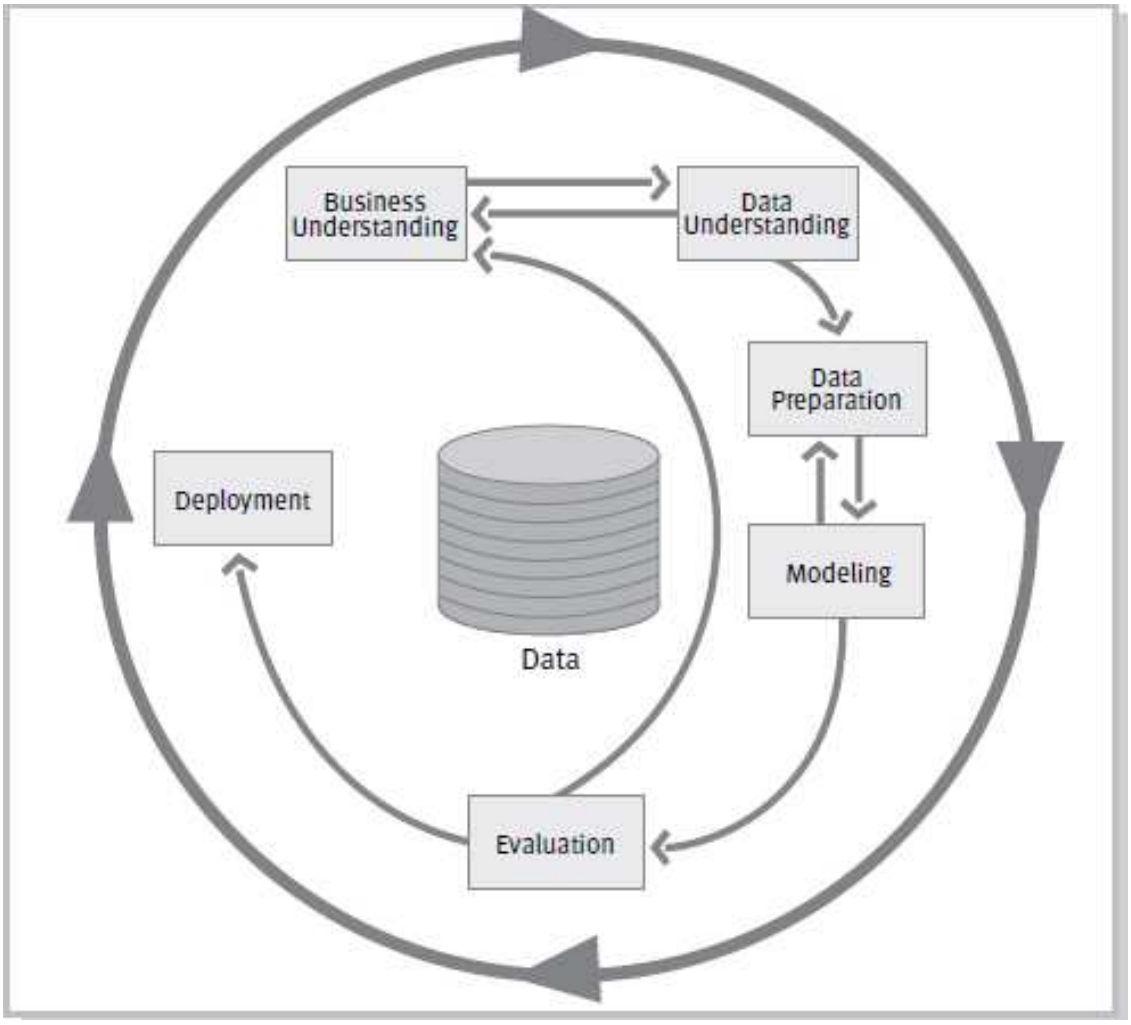


Figure 4: Phases of the CRISP-DM reference model

The CRISP-DM technique is described as a hierarchical process model, with four levels of abstraction (from general to specific): phase, generic task, specialized task, and process instance (see figure 4). At the top level, the data mining process is organized into several phases; each phase consists of several second-level generic tasks. This second level is generic, and it is intended to be wide-ranging enough to cover all data mining situations. The generic tasks are intended to be as complete and stable as possible. Complete means covering

both the whole process of data mining and all data mining applications. Stable means that the model should be valid for yet unforeseen developments like new modeling techniques.

The third level, specialized task level, is where you explain how to carry out general task activities in unique situations. At the second level, for example, a general task called clean data may be found. The third level delves into how this task differs based on the situation, such as whether the problem is cleaning numeric or categorical variables, or whether the challenge is clustering or predictive modeling.

The representation of phases and tasks as discrete stages completed in a precise order represents an idealized sequence of occurrences. Many of the jobs, in fact, can be accomplished in any order, and it will be necessary to return to previous tasks and repeat specific procedures numerous times. Because capturing all of these possible routes through the data mining process would necessitate an unduly complex process model, our process model does not attempt to do so.

The process instance, the fourth level, is a record of the actions, decisions, and outcomes of a real-world data mining project. A process instance is arranged according to the higher-level tasks, but it depicts what happened in a specific engagement rather than what happens in general.

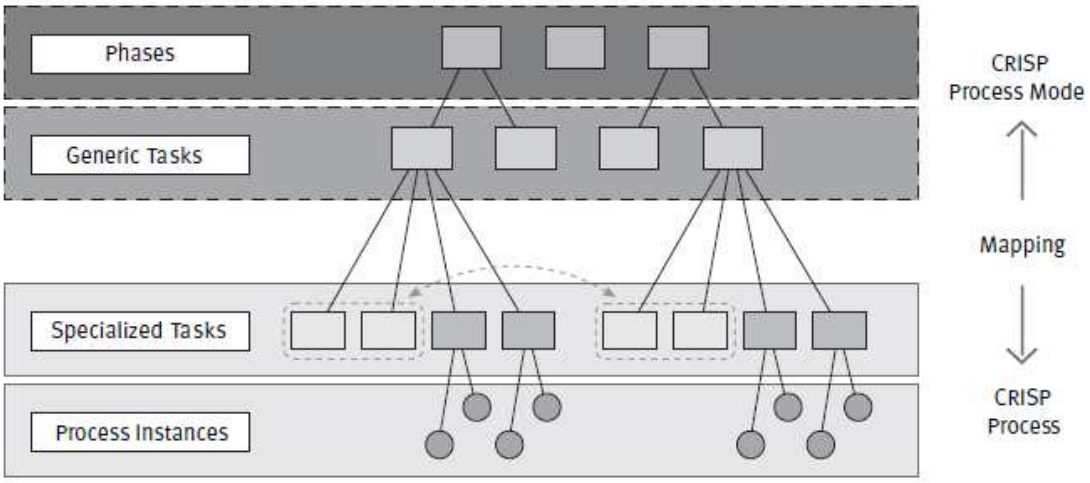


Figure 5: Four level breakdown of the CRISP-DM methodology (SPSS)

CRISP-DM with its six main steps and a total of 24 tasks and outputs, is more refined as compared to KDD. The 24 tasks and output are depicted in Table 1 below:

Business Understanding	Data Understanding	Data Preparation	Modeling	Evaluation	Deployment
Determine Business Objectives *Background* *Business Objectives* *Business Success Criteria* **Assess Situation** *Inventory of Resources* *Requirements, Assumptions, and Constraints* *Risks and Contingencies* *Terminology* *Costs and Benefits* **Determine Data Mining Goals** *Data Mining Goals* *Data Mining Success Criteria* **Produce Project Plan** *Project Plan* *Initial Assessment of Tools and Techniques*	**Collect Initial Data** *Initial Data Collection Report* **Describe Data** *Data Description Report* **Explore Data** *Data Exploration Report* **Verify Data Quality** *Data Quality Report*	**Select Data** *Rationale for Inclusion/ Exclusion* **Clean Data** *Data Cleaning Report* **Construct Data** *Derived Attributes* *Generated Records* **Integrate Data** *Merged Data* **Format Data** *Reformatted Data* *Dataset* *Dataset Description*	**Select Modeling Techniques** *Modeling Technique* *Modeling Assumptions* **Generate Test Design** *Test Design* **Build Model** *Parameter Settings* *Models* *Model Descriptions* **Assess Model** *Model Assessment* *Revised Parameter Settings*	**Evaluate Results** *Assessment of Data Mining Results w.r.t. Business Success Criteria* *Approved Models* **Review Process** *Review of Process* **Determine Next Steps** *List of Possible Actions* *Decision*	**Plan Deployment** *Deployment Plan* **Plan Monitoring and Maintenance** *Monitoring and Maintenance Plan* **Produce Final Report** *Final Report* *Final Presentation* **Review Project** *Experience Documentation*

Table 1: Generic tasks (bold) and outputs (italic) of the CRISP-DM reference model (SPSS)

A variety of adjustments and extensions of the CRISP-DM approach have been proposed, similar to the KDD methodology, with the two primary directions. —extensions of the process model itself and adaptations, merger with the process models and methodologies in other domains Plotnikova, V., Dumas, M., Milani, F., (2020). Extensions direction of process models to integrate both Data Mining & Knowledge Discovery (DMKD) process model Cios, K., & Kurgan, L., (2005). It includes numerous explicit feedback mechanisms, a revision of the last step to incorporate newly obtained knowledge and insights, and it relies on technology to deploy results. Moyle, S., & Jorge, A., (2001), Blockeel, H., Moyle, S., (2002) proposed Rapid Collaborative Data Mining System (RAMSYS) framework—this is both data mining methodology and system for remote collaborative data mining projects. The RAMSYS attempted to achieve the combination of a problem solving methodology, knowledge sharing, and ease of communication Plotnikova, V., Dumas, M., & Milani, F., (2020). It was designed to enable for the disciplined work of remotely located data miners in terms of information flow while allowing for the free flow of problem-solving ideas. S. Moyle, A. Jorge, S. Moyle, S. Moyle, S. Moyle, S. Mo (2001). Industrial Engineering by Solarte, J., (2002), and Software Engineering by Marbán, O., et al. (2007, 2009) presented CRISP-DM adaptations and integrations with other particular fields. Both techniques improved CRISP-DM by adding new phases, activities, and tasks that are common in engineering

processes, such as ongoing support Solarte, J., (2002), project management, organizational, and quality assurance tasks Marbán et al (2009).

After CRISP-DM was created, a small number of attempts to construct independent or semi-dependent data mining frameworks were made. The KDD Roadmap by Debuse, J., et al. (2001) for a proprietary prediction toolset (Lanner Group) and IBM's current effort with Analytics Solutions Unified Method for Data Mining (ASUM-DM) in 2015 were both driven by industry participants. Additional duties were contributed by both frameworks, such as resourcing in KDD Roadmap or a hybrid approach adopted in ASUM, such as a combination of agile and traditional implementation methods. The evaluated data mining process models and approaches are summarized in Table 2 by their origin, premise, and essential principles.

Name	Origin	Basis	Key concept	Year
Human-Centered	Academy	KDD	Iterative process and interactivity (user's point of view and needed decisions)	1996, 2004
Cabena et al.	Academy	KDD	Focus on data processing and discovery tasks	1997
Anand and Buchner	Academy	KDD	Supplementary steps and integration of web-mining	1998, 1999
Two Crows	Industry	KDD	Modified definitions of steps	1998
SEMMA	Industry	KDD	Tool-specific (SAS Institute), elimination of some steps	2005
5 A's	Industry	Independent	Supplementary steps	2003
6 Sigmas	Industry	Independent	Six Sigma quality improvement paradigm in conjunction with DMAIC performance improvement model	2003
CRISP-DM	Joint industry and academy	KDD	Iterative execution of steps, significant refinements to tasks and outputs	2000
Cios et al.	Academy	Crisp-DM	Integration of data mining and knowledge discovery, feedback mechanisms, usage of received insights supported by technologies	2005
RAMSYS	Academy	Crisp-DM	Integration of collaborative work aspects	2001–2002
DMIE	Academy	Crisp-DM	Integration and adaptation to Industrial Engineering domain	2001
Marban	Academy	Crisp-DM	Integration and adaptation to Software Engineering domain	2007
KDD roadmap	Joint industry and academy	Independent	Tool-specific, resourcing task	2001
ASUM	Industry	Crisp-DM	Tool-specific, combination of traditional Crisp-DM and agile implementation approach	2015

Table 2: *Key aspects of existing data mining process models and methodologies Plotnikova V, Dumas M, Milani F. (2020)*

2.3 Data Mining Techniques

Data Mining Techniques (DMT) represent one of the three core areas of competency that are needed to be successful in DM and is regarded as one of the three pillars of DM Berry, M., & Linoff, G., (2000) and it provide a popular and powerful tool set to generate various data driven classification systems Churilov, L. et al., (2005) and has been shown to play an important role in rule refinement even if the sample size is limited Hilage, T., & Kulkarni, R., (2012). Data mining adopt its technique from many research areas, including statistics machine learning, database systems, rough sets, visualization and neural networks.

2.3.1 Statistics

Statistics includes several methods to analyze numerical data in large quantities. Regression analysis, cluster analysis, correlation analysis, and Bayesian networks are some of the statistical tools used in data mining. Statistical models are usually built from a training data set. Correlation analysis identifies the correlation of variables to each other. Bayesian network is a directed graph that represents causal relationship among data found out using the Bayesian probability theorem.

2.3.2 Machine Learning

Machine learning is a set of methodologies, ideas, and algorithms that allows for learning and prediction based on historical data. Machine learning is used to build new models and to search for a best model matching the test data. Machine learning methods normally use heuristics while searching for the model. The most common machine learning methods used for data mining are inductive concept learning, conceptual clustering and decision tree induction Gheware, S., et al., (2014). A decision tree is a classification tree that decides the class of an object by following the path from the root to a leaf node and they are induced by the training set and classification rules.

2.3.3 Database Oriented Techniques

Advancements in database and data warehouse implementation helps data mining in a number of ways. Database-oriented strategies are mostly used to build data characteristics Some of the database focused approaches extensively used in data mining are iterative database scanning for frequent item sets, attribute focusing, and attribute-oriented induction. Iterative database scanning searches a database for frequently occurring item sets. Using conceptual hierarchies, attribute-oriented induction generalizes low-level facts into high-level concepts.

2.3.4 Neural Networks

Information processing is represented by neural networks, which are a brain metaphor. Rather than being a precise reproduction of how the brain works, these models are scientifically inspired. Neural networks have been shown to be very promising systems in many forecasting applications and business classification applications due to their ability to "learn" from the data, their nonparametric nature (i.e., no rigid assumptions), and their ability to generalize. Neural computing is a machine learning pattern recognition technology. An artificial neural network is the name given to the model that emerges from neural computing (ANN). ANN have been used in many business applications for pattern recognition, forecasting, prediction, and classification. An artificial neural network (ANN) is made up of neurons, which are connected nodes. A neuron is a computational device that uses its inputs to compute some requirement; the inputs can even be the outputs of other neurons. By altering the connections and parameters of the nodes, an ANN may be taught to determine the link between input and output qualities.

2.3.5 Data Visualization

The information gathered from vast amounts of data should be presented to the end user in a clear and understandable manner, which data visualization techniques enable. Data is transformed into different visual objects such as dots, lines, shapes etc and displayed in a two- or three-dimensional space. Data

visualization is an effective way to identify trends, patterns, correlations and outliers from large amounts of data.

2.4 Data Mining Task

Data mining tasks can be either directed or undirected. Most of the data mining task in the real world are directed Berry, M., & Linoff, G., (2000). In directed data mining the goal is to predict, estimate, classify, or characterize the behavior of some pre-identified target variable of variables in terms of a collection of input variables. Examples of directed data mining include classifying insurance claims, estimating loan balances, identifying prospects, and predicting attrition. In undirected data mining, there is no target variable to be predicted. Instead, the goal is to discover structure in the data set as a whole. Determine what products should be put together for a speciality catalog, uncover groups of readers or listeners with similar likes in books and music, and discover natural consumer segmentation for market analysis are all examples of undirected data mining.

2.4.1 Tracking patterns

Learning to discover patterns in your data sets is one of the most basic data mining approaches. This is frequently an identification of a periodic aberration in your data, or the ebb and flow of a particular variable through time. It entails detecting and tracking trends or patterns in data in order to draw educated conclusions regarding business outcomes. When a company notices a pattern

in sales data, for example, there's a foundation for taking action to capitalize on the information.

2.4.2 Classification

It is one of the most used techniques, to develop models that can population records at large. Data classification is a process in which the given data is classified into different classes according to a classification model. This technique is employed in the classifier training algorithm for company development. Various classifiers used for the classification algorithms such as Decision tree, Bayesian classifier, ANN, Support vector machine etc. The two types of classification are:

- Supervised classification
- Unsupervised classification

2.4.3 Clustering

Clustering is the process of grouping data into multiple distinct classes to better describe the data. It divides a large data collection into smaller groups to simplify the design and implementation process. Clustering's goal is to maximize the similarity between class objects while reducing the similarity between classes. It is used to identify objects that belong to the same class. Types of Clustering are

- Partitioning methods

- Hierarchical agglomerative methods
- Density based methods
- Grid based methods
- Model based methods

2.4.4 Outlier detection

Outlier detection determines any anomalies in datasets. We need to be able to identify anomalies and outliers in our data. Once organizations find aberrations in their data, it becomes easier to understand and replicates why these anomalies happen and prepare for any future occurrences to best achieve business objectives.

2.4.5 Regression

It can be used to make forecasts. The link between one or more independent and dependent variables is modeled using regression analysis. This process is similar to classification the major difference is that the object to be predicted is continuous rather than discrete. Types of regression includes:

- Linear regression
- Non-linear regression
- Multi-variate linear regression
- Multi-variate non-linear regression

2.4.6 Association

In this process the association between the objects is found. It discovers the association between various data and the association between the attributes of these data. Association also called Affinity Grouping is used to determine which things go together. Types of Association rules are:

- Multi-variate association rule
- Multi-dimensional association rule

2.4.7 Summarization

With the use of visualization tools, this report creation technique allows for better decision making for big volumes of client data. It will make business decision-making more functional. It's fantastic for data mining. Through using visual tools, analysts can reach a better understanding of the data because they can focus their attention on some of the patterns found by other method Solarte, J., (2002).

2.4.8 Prediction

Prediction is a very powerful and most valuable data mining techniques, and it is the bedrock of predictive analytics. Prediction uses patterns found in current and historical data to extend them to accurately predict what would happen in the future. There are several different approaches to using predictive analytics. Aspects of machine learning and artificial intelligence are used in

some of the more advanced ones. Predictive analytics, on the other hand, does not have to rely on these techniques; it can also be aided by simpler algorithms.

2.5 Types of Data Mining Systems

There are a plethora of data mining solutions on the market or in development. Some are dedicated to a certain data source or have limited data mining capabilities, while others are more versatile and extensive. Data mining systems can be categorized according to various criteria (Zaïane, 1999)

Classification according to the type of data source mined: this classification categorizes data mining systems according to the type of data handled such as spatial data, multimedia data, time-series data, text data, World Wide Web, etc.

Classification according to the data model drawn on: Data mining systems are classified according to the data model they use, such as relational databases, object-oriented databases, data warehouses, transactional databases, and so on.

Classification according to the kind of knowledge discovered: This categorization divides data mining systems into categories based on the type of knowledge obtained or data mining capabilities, such as characterization, discrimination, association, classification, clustering, and so on. Some systems are multi-functional data mining systems.

Classification according to mining techniques used: Data mining systems employ and provide different techniques. This classification categorizes data mining systems according to the data analysis approach used such as machine

learning, ANN, genetic algorithms, statistics, visualization, database oriented or data warehouse-oriented, etc. The classification can also take into account the degree of user interaction involved in the data mining process such as query-driven systems, interactive exploratory systems, or autonomous systems Zaïane, O., (1999).

2.6 Data Mining application in Insurance

Data Mining was originally developed to act as expert system to discover problems and did not require assumption to be made about data. Nowadays companies and individuals apply data mining in order to achieve their goals. Data mining is becoming more popular in both the private and public insurance industries. Customer information is one of a company's most valuable assets. Traditional approaches for dealing with massive amounts of data generated by insurance transactions are far too complicated. Data mining is a way for transmitting large amounts of data for decision-making.

Insurer employs data mining approaches to improve research and promote client sales. The following are some of the jobs that data mining is utilized for in the insurance industry. K. Umamaheswari and D. Janakiraman (2014).

2.6.1 Acquiring new customers

Acquisition of new customer is most important scenario of any firm. Traditionally, the insurance companies used the services of brokers and agents to acquire the customers, but today a lot of ways helps to acquire the new

customers. The insurance company focuses on recruiting new customers as well as retaining existing ones. Cluster Analysis used in the private sector to identify target group of customers. It involves targeting the population who are most likely to become customers or most profitable to the company.

2.6.2 Customer level analysis

It is analysis of customer purchase patterns and behavior. Using associated discovery technique, most insurance firms accurately select which policies and services to offer which customers. Data mining technology was used for insurance settlement, and it was used to evaluate customer information and construct a function structure model for customer analysis..

2.6.3 Customer Segmentation

Segment based products for targeting the customers. Data mining can be used to categorize customers, encourage cross-selling of services, and improve customer retention. Based on their purchase history, customers are assigned to a lifestyle segment. Market segmentation is critical for the creation of long-term consumer connections. V. Miguéis et al., (2012).

2.6.4 Policy designing and policy selection

The insurance company looked at whether consumers buy policies for the right reasons and for the right policies. In that situation, insurance companies employed data mining tools to compete successfully in the market.

2.6.5 Prediction

Data mining used for variety of applications such as predicting and classifying customer's and clustering customer characteristics for achievement of profitability. Predictive analytics has various advantages for customers, including as a simplified claim handling process, lower policy premiums for low-risk customers, and speedier and automated claim payment. Data mining techniques can forecast future behaviors and trends, allowing firms to make proactive, data-driven decisions. It performs inference on the current data (insurance dataset) order to make predictions Balaji, S., & Srinivasta, K., (2012)

2.6.6 Claims management

Data mining handled claims management functions such as claim analysis and fraud analysis, which is one of the most important functions in the insurance industry. Creating new product lines The development of a new product or plan is dependent on the needs of the consumer. Insurance companies used all of their accessible data to better build new products and marketing efforts, according to Umamaheswari & Janakiraman (2014).

2.6.7 Underwriting and Policy management

In this application, data mining can be used to optimize the function of the insurance value chain (Premium Analysis and Loss analysis) Data mining can assist insurance businesses in developing a solid loss prevention strategy that allows them to assess and manage risks..

2.6.8 Risk management

Insurance is, of course, one of the most important elements in the risk management process. The insurance sector is very concerned about recognizing the risks that affect their business. The insurance industry is moving towards risk-based practices all across the world (capital, supervision, pricing). Risk-based pricing, on the other hand, sets prices to reflect risk, cover operational and compliance costs, and provide a long-term profit given the business's nature. Data mining could aid in achieving the delicate balancing act that is required for risk management.

2.6.9 Reinsurance:

In the field of risk management, reinsurance is included. Data mining tools can develop predictive models to arrive at the retention limits of the insurance company for each line of business based on the historical claims data. Based on the loss experience of similar policies in the past, these predictive algorithms can identify acceptable policies for reinsurance.

2.6.10 Fraud detection

Detecting fraud claims is important in the insurance firm. Data mining isolates the factors that lead to fraud waste and abuse. To figure out which transactions are the most likely to be fake.. This is called as Fraud anomaly detection.

And he discussed the characteristics of fraud detection are high claims payment data is incorrect, suspicious data analysis, problem of hospital or physician Lin,

K., & Yeh, C. (2012). The types of fraud and how much of fraud activities in the insurance firm discussed and he developed a claim sorting algorithm for the claim processing systems Derrig, R., (2002).

2.6.11 Trend analysis

The discipline of evaluating changes in social patterns, such as fashion, technology, and consumer behavior, is known as trend analysis. Trend analysis also Applicable in insurance to reveal difference between the typical customers of this month and last month Venkatesh, M., (2013). Data mining used in the different service industry especially in insurance firms the most frequently used applications for customer segmentation, customer retention, risk assessment and fraud detection and Policy approval process.

2.7 Related works

In Nigeria, there are works done to assess the application of DM in the different sectors like education, airlines, banking, telecommunication, insurance, health care, and customs.

In the educational sector, Adekitan, et al (2019a) conducted a research on DM approach for predicting the daily Internet data traffic of a smart university while Adekitan, et al (2019b) used DM to determine the performance of first year student using their admission requirement, further to this, Adekitan, et al (2020) conducted a research on how ethnicity play a role in improving learning process using DM.

In the telecommunication sector, Eze U et al., (2017) conducted a research on the application of DM to the telecommunication industry in Nigeria and how it can be used for target marketing, how best to address problems of voice and data quality and how to attract, retain and manage customer.

In government sector, Obunadike, G., & Ume, A., (2014) in their research examine how DM can be used with data warehouse to improve prison service and reduce crime in Nigeria, Ujah, B., & Adejoro, C., (2016) in their study highlighted how DM can be use to generate investigative lead and intelligence to combat crime and further recommend a framework for proper intelligence gathering and information sharing between security operative.

In the health sector Uba, M., eta al., (2019) in their study used DM to build, developed and validate a comprehensive model for predicting predicting diabetes mellitus and other chronic diseases in north-western part of Nigeria. Peter, I., et al., (2013) conducted a research using DM to predict Immunize-able diseases in Nigeria.

Many researchers applied different DM techniques and algorithms for insurance use case. Guo, L., (2003) also applied DM to model insurance risk in the property/casualty insurance. Rejesus, M., et al. (2004) conducted a research to study the application of DM to detect fraud in crop insurance in US. In addition to this, Koh, C., & Gervais, G., (2010) applied DM techniques and algorithms for detecting motor insurance fraud.

In Africa, Mesfin, F., (2005) researched on the possible application of predictive DM techniques in the renewal process of personal accident policies, Tesfaye (2002), developed a predictive model using DM techniques in support of insurance risk assessment. Tariku, A., (2011) researched on using DM in detecting insurance fraud in the insurance industry in Ethiopia.

In Nigeria financial services industry, most research in DM was concentrated in banking and stock market, very little research exist in DM around insurance the only exception is in health insurance where for instance Yange, S., (2019) conducted a research using DM to detect fraudulent activities by all actor of NHIS.

To the knowledge of the researcher there are only few attempts in our country that have been done so far towards the application of DM in the insurance industry all of which are Health Insurance. But till now there is not any work that have been done so far regarding the application of DM for the insurance industry that amongst others can handle the three critical areas of proper risk assessment, fraud detection and having analytical CRM. Hence this study has a great contribution in applying DM technology in the insurance industry.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

The insurance rate used to determine the premium and proper pricing of a life policy is determined by age, gender, coverage amount, coverage type, tobacco use, family history, occupation and mortality table, the ability to predict or classify customers risk level is very important and non-trivial. In addition, fraudulent claims are an ever-present, ever-increasing problem for insurer, in the United States alone FBI estimates that the total cost of insurance fraud (non-health insurance) is estimated to be more than $40 billion per year. The techniques for identifying and mitigating fraud are critical for the long-term success of insurance firms. To be able to conduct proper risk assessment, detect fraud and gain competitive advantage, there is need to have a analytical CRM strategy in place. These problem space required a three-prong approach which can be solved with DM. The three processes that would form the basis of our proposed model are:

- Risk management process
- Fraud detection process and
- Analytical CRM process.

The risk management process entails applying policies, procedures, and practices to activities such as communicating and consulting, setting the context, and assessing, treating, monitoring, reviewing, recording, and reporting risk (ISO, 2018). This risk management design process is depicted in Figure 7 and its five steps are detailed below.

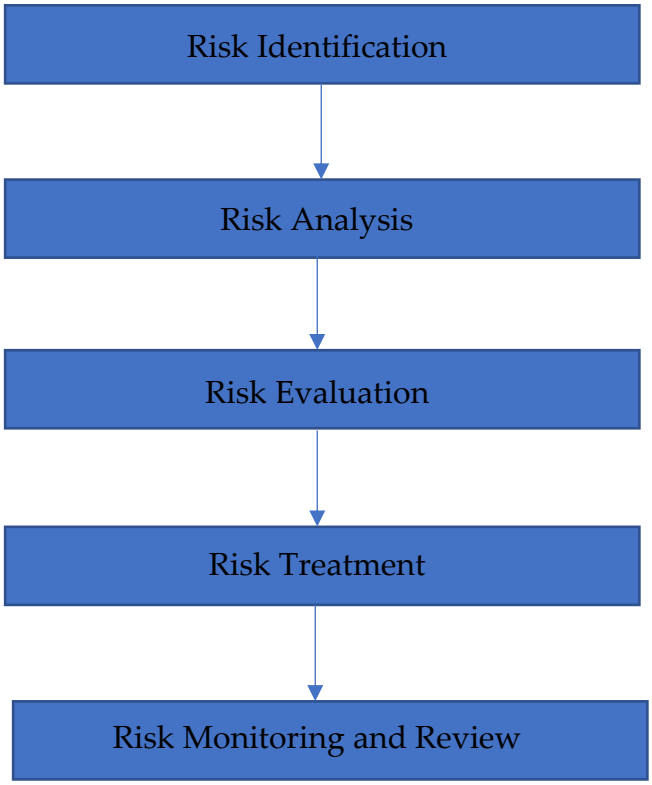


Figure 6: Risk Management Process

The fraud detection design process is depicted in Figure 8 and its three steps are detailed below.

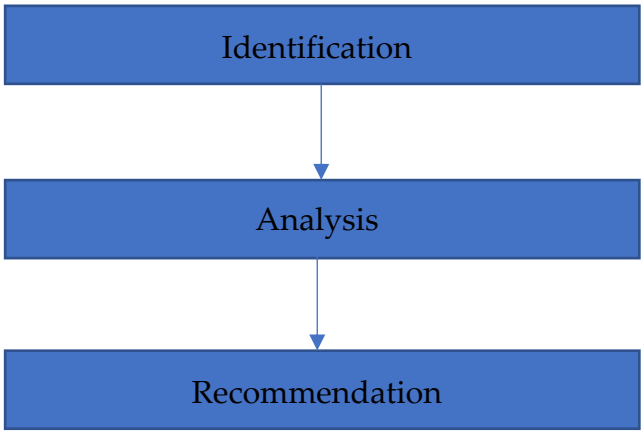


Figure 7: Fraud detection process

The analytical CRM strategy process also know as customer discovery process is depicted in Figure 9 and its three steps are detailed below.

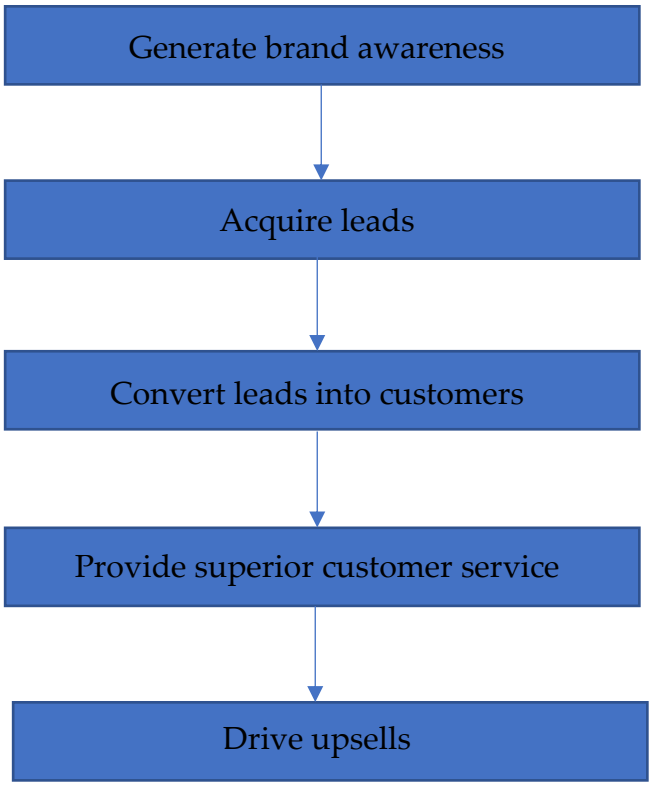


Figure 8: Analytical CRM strategy process

We would use the engineering process as described by Landis, R., (2013), in the execution of this study. This modified engineering design process is depicted in Figure 10 and its seven steps are detailed below.

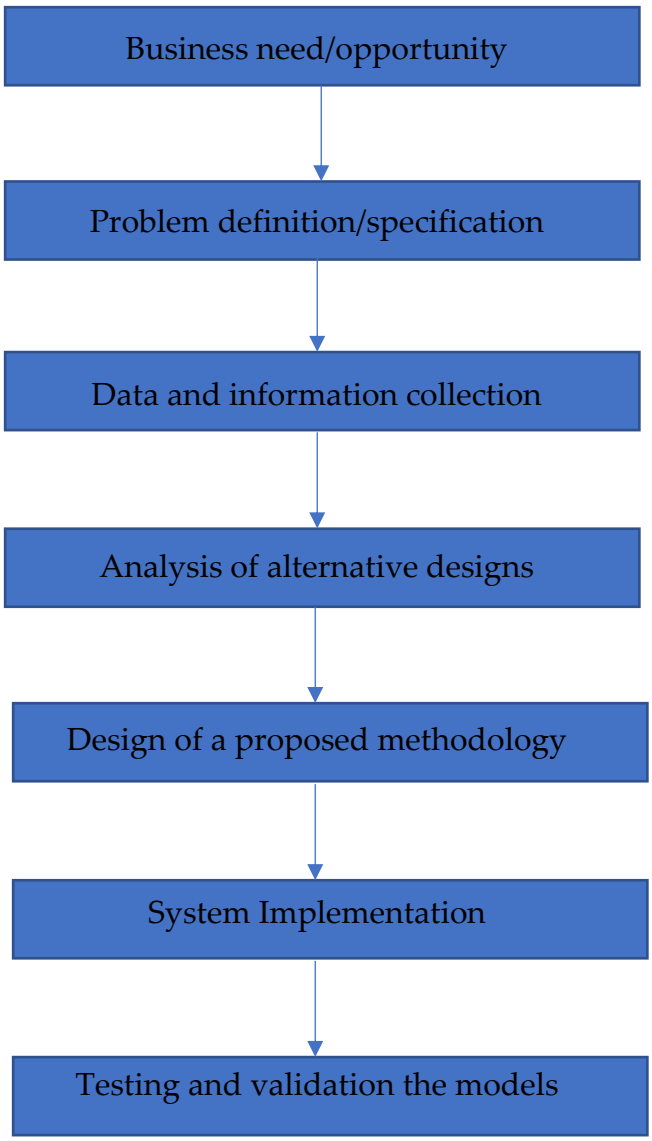


Figure 9: The applied engineering design process

3.2 Business need/opportunity

The first step in solving a business problem is the identification of a need or opportunity. For insurers, these needs, and opportunities are immense and

diverse. Insurers have wide range of services namely: term life, whole life, endowment, unit-linked, group life, moneyback, saving and investment linked, retirement insurance and child insurance. Various products and customization are made on these services to cater for the market niche. The insurance product development processes by Society of Actuaries (2017) are:

1. **Idea Generation:** To find candidates for formal analysis, new product or feature ideas are assessed and screened.

2. **Product Concept and High-Level Feasibility:** Each candidate proposal is given a business case or a feasibility study. Selected ideas are organized into projects, which are then sponsored.

3. **Product Planning and Design:** Products and features are designed.

4. **Establish Underwriting Guidelines:** Underwriting rules and processes are established, with an appropriate classification of risks.

5. **Product Pricing:** Actuarial pricing that includes setting the rates, reserves, compensation, and any other pertinent data. This data is stress tested against product profit objectives and expected or actual data.

6. **Reinsurance:** Pricing and treaty negotiation

7. **Update IT Systems Day 1:** For each gap, system requirements are defined and implemented to cover both what a client can do

contractually on Day 1 and what your business requires on Day 1. Illustration systems are included.

8. **Update IT Systems Day 2:** Any further functionalities and organizational requirements not required Day 1 are designed and implemented as systems requirements.
9. **Update Business Procedures:** For each gap, operations and business procedures are built.
10. **Marketing Plans:** Materials and campaigns for marketing are created and implemented.
11. **State Filings:** Forms for products are created and filed. This provides anything from filing informational products (no approval required) to full forms/rates filing.
12. **Product Launch:** Product Development Complete
13. **Tracking of Results vs. Plan/Business Case:** comparing actual business results to pricing/plan to gauge assumptions versus experience

Much of this information is available; the challenge is getting to it. The ideal tool to assist in that effort is DM.

DM can help insurers process and analyze information, deciding on the most effective DM techniques and systems can be complicated. Like DM like insurance is the result of the confluence of multiple disciplines and for that

reason the process of implementing DM process in insurance is difficult and requires much specialized knowledge.

3.3 Problem Definition

There are so many options, tasks, techniques, tools, formats, and approaches to risk assessment, fraud detection and analytical CRM that insurer would find very difficult to design and implement. Methodologies already exist but the organizational context of each insurer is different as such there is no clarity as to the approach of DM best suited for obtaining the vast array of data needed by insurer to tackle the problem associated with risk assessment, fraud detection and analytical CRM. Thus, a DM methodology to meet the specific requirements of insurer problem areas is needed. The methodology would help insurer in selecting appropriate DM tools to do proper risk assessment, minimize fraud and have analytical CRM projects from a systems perspective.

3.4 Data and Information collection

To accomplish this study, analysis, reviews, and comparisons of DM methodologies and applications were conducted. These were based on vendors information various research works, and case studies available in literature and research publications.

3.5 Analysis of alternatives

There are several different DM methodologies, but there is no one standard methodology for applying DM. As a result, a number of companies have developed their own proprietary methodology. There are various disadvantages to these. Approaches developed by software providers are highly associated with the design of their own solutions and software packages. DM however is considered a sort of art where the miner can choose the best path to get the job do rather than follow the strict procedure of the vendor. Individual preferences and intuition are extremely important to DM in insurance domain this does not in any way negate following all the essential steps of DM methodologies to obtain superior results.

The most popular methodologies are KDD, SEMMA and CRISP-DM. The next sections go through them in detail.

3.5.1 KDD

The KDD process, as presented in Fayyad, U., et al, (1996a) is the process of using DM methods to extract what is deemed knowledge according to the specification of measures and thresholds, using a database, as well as any necessary preprocessing, subsampling, and database transformation. There are considered five stages, presented in figure 1:

1. **Selection:** This stage consists of creating a target data set by focusing on a subset of variables or data samples, on which discovery is to be performed.

2. **Pre-processing:** This stage consists of cleaning the target data in order to obtain consistent data.

3. **Transformation:** This stage consists of the transformation of the data using dimensionality reduction or transformation methods.

4. **Data Mining:** This stage consists of the searching for patterns of interest in a particular representational form, depending on the DM objective (usually, prediction)

5. **Interpretation/Evaluation:** This stage consists on the interpretation and evaluation of the mined patterns.

The KDD method is interactive and iterative, requiring the user to make several decisions over the course of several steps. Brachman, R., & Anand, T., (1996). However, it fails to address the critical problems of understanding the business, the application domain, relevant knowledge and inputs to understating the goals of the end-user and how to consolidate the knowledge into the system Fayyad, U., et al, (1996a).

3.5.2 SEMMA

The SEMMA process is a DM methodology proposed and developed by the SAS Institute. SEMMA is for Sample, Explore, Modify, Model, and Assess, and it relates to the steps involved in performing a DM project. SEMMA offers a DM process that has a cycle with 5 stages:

1. **Sample:** This stage consists of sampling the data by extracting a small portion of a large data set that have significant information, yet small enough to manipulate quickly.

2. **Explore:** This stage consists of exploring the data by searching for trends and anomalies with the purpose of gaining understanding and ideas about the data.

3. **Modify**: This stage consists of modifying the data by creating, selecting, and transforming the variables to focus the model selection process.

4. **Model:** This stage consists of modeling the data by allowing the software to search automatically for a combination of data that reliably predicts a desired results.

5. **Assess:** This stage consists of assessing the data by evaluating the usefulness and reliability of the findings from the DM process and estimate how well it performs.

Although the SEMMA process is independent from the DM chosen tool, it is linked to the SAS Enterprise Miner software and cannot be applied outside the system, it only pretends to instruct users on how to use the DM applications. It concerns only the statistical, modeling, and the data manipulation parts of the data-mining process. It is missing some of the most important components of any information systems project, such as the analysis, design, and implementation phases.

The roles of the organization and stakeholders are not considered in the SEMMA methodology during the project, and data mining is not seen as an integrated element of the entire system and company..

3.5.3 CRISP-DM

The CRISP-DM method was created via the efforts of a group of DM providers and enterprises, which included DaimlerChrysler, SPSS, and NCR at the outset. CRISP-DM stands for CRoss-Industry Standard Process for DM. It was conceived in 1996 but was not completed until 1999. It's designed to work with any industry tool or application.. It consists of the following stages:

1. **Business understanding:** This initial phase focuses on understanding the project objectives and requirements from a business perspective, then converting this knowledge into a DM problem definition and a preliminary plan designed to achieve the objectives.

2. **Data understanding:** The data understanding phase starts with an initial data collection and proceeds with activities to get familiar with the data, to identify data quality problems, to discover first insights into the data or to detect interesting subsets to form hypotheses for hidden information.

3. **Data preparation:** The data preparation phase covers all activities to construct the final dataset from the initial raw data.

4. **Modeling:** In this phase, various modeling techniques are selected and applied, and their parameters are calibrated to optimal values.

5. **Evaluation:** At this stage, the model (or models) obtained are more thoroughly evaluated and the steps executed to construct the model are reviewed to be certain it properly achieves the business objectives.

6. **Deployment:** Creation of the model is not the end of the project. Even if the purpose of the model is to increase knowledge of the data, the knowledge gained will need to be organized and presented in a way that the customer can use it. Chapman, P., et al., (2000)

CRISP-DM approach has broader applications than SEMMA, one of its major drawbacks is that it combines tools (software packets) and techniques in the same category. Selecting the tools and techniques simultaneously introduces a bias towards using a specific data mining tools which may not be the most

relevant to the purpose of the study and may not be needed for that purpose. This may also cause the study to be biased by under-analyzing the organization's aims and requirements. In data mining projects, it is important to analyze the organization's needs, requirements, goals, and strategies in a neutral manner. For example, organizations may need to extract information from their data warehouses on a one-time or recurring basis.

Techniques covered by CRISP-DM may be used because they are included in the organization's tools, rather than because they are truly required As a result, the outcomes of this approach may not adequately match to the organization's key objectives, and the models developed in this manner may not accurately represent the behavior of the entities for which the study was originally intended. This may specificity be true for insurance usage such as risk management, fraud detection, analytical CRM and many more; but they are not limited to insurance alone and can be applied to many different data mining projects in other fields.

Despite the above, CRISP-DM methodology is useful. It describes a data-mining project as a six-phase cycle with no fixed order of phases. The first phase very important elements of DM such that the business's objectives, requirements, constraints, and resources available for the project, to establish that the data mining goals is fully understood and captured.

CRISP-DM consider collecting and analyzing available data and assumes the information is readily available; this is not always the case, new information available during the data preparation stage can have significant effect on the model to be used and may not guaranteed to provide solution to the problems of the organization.

Techniques used for DM should be based on the organization's goals and requirements per time and not on available data. The availability of new data must result in the revision of all assumptions and such changes would result in making new assumption that would render an assumption once valid to be incorrect.

The CRISP-DM methodology and other techniques similarly emphasize the need to partition data into training and validation sets depending on the technique used. Following the construction of the models with the training data, a validation test is done to confirm that the model created behaves as closely as possible to the real system. Asides from the difficulties CRISP-DM methodology presents, it is a good approach to DM and the DM cycle.

3.6 Design of a proposed methodology

The three two major DM methodologies are useful in their ways and have wide range usage, they may not be the most useful methodologies for broad usage in insurance. Through the revision of the DM methodologies discussed in the previous section and the application of information systems analysis and

design structure, a proposed methodology for the application of data mining in life insurance is described in the next chapter.

CHAPTER 4

A PROPOSED METHODOLOGY

4.1 Introduction

DM methodology must follow a structured approach to problem-solving. This would make it possible to replicate the methodology and apply it to many domains. DM should not be a one-off implementation; the researcher believes it should be ingrained in the organizations core and operational processes. This would go a long way in reaping values for the initiatives. Such would further assist in determining where errors have occurred in the process and give assurance on the result and recommendations.

The value creation of the proposed DM methodology is based on these principles:

1. **Integrated:** DM must be an integral part of all organizational activities.
2. **Structured and wide-ranging:** DM must follow a structured approach and adopt a wide-ranging approach to have a consistent and comparable results.
3. **Customized:** DM framework and process must be highly customizable and vendor neutral as such it would be easily adaptable to the organization's external and internal context.

4. **Inclusive:** DM thrives on data as such there is need to involve a stakeholder from the onset and capture experience, knowledge, and views.

5. **Dynamic:** The business environment is not static as such DM must be able to anticipates and respond to the changing business environment with agility.

6. **Risk-based:** DM must put into consideration the various risk associated with the initiatives and strive to eliminate all forms limitations and uncertainties.

7. **Environmental and Human factors:** The environment and culture of the organisation must be known; the human factor must be determined so as to remove all form of bias and eliminate any influence on the result.

8. **Continual improvement:** DM must put measure in place to improve learning and continually improve the result.

It is premised on these principles and after extensive review of various DM methodology that the researcher provides a framework for using data mining in solving problems related to insurance and one that can be applied to other domains. The proposed structured approach would ensure data miners obtain all the required information for solving the problem. We would look at it from

a process perspective without bias for any tools. The methodology and processes are:

1. Organisational Analysis
2. Project and Change Management
3. Data Mining
4. Interpretation and Validation
5. Presentation and Validation
6. Integration, Monitoring and Review

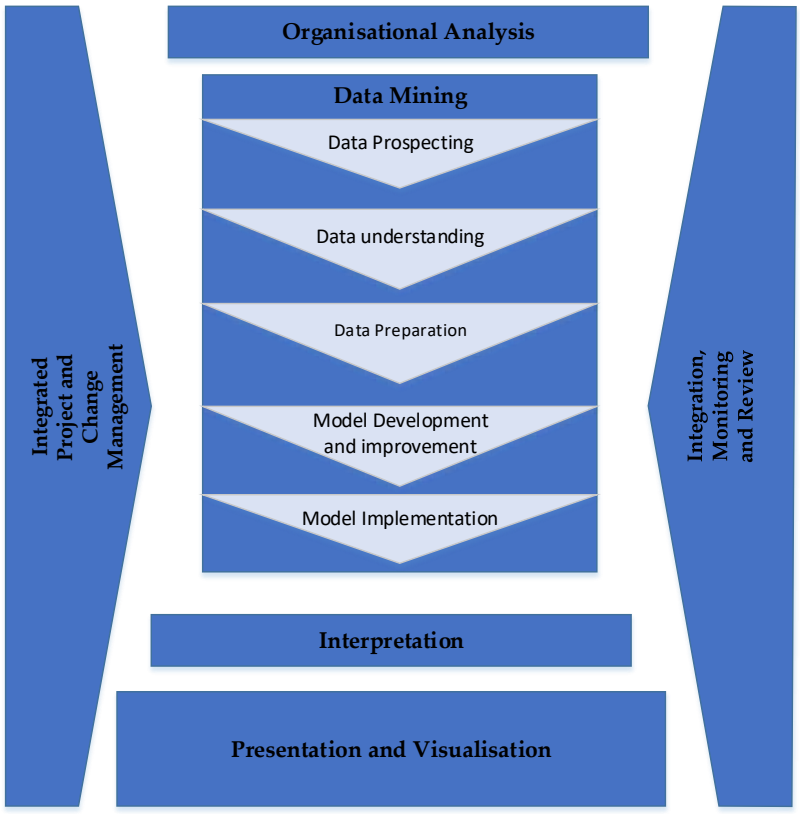


Figure 10: Proposed DM Methodology

These steps are shown in Figure 4.1 and described in detail in the following sections.

4.2 Organisational Analysis

The first and the most important step for any DM project is to understand the existence of the organisation and the DM project itself. DM cannot be done in isolation of the organisation, it must examine and understand the organization's mission, vision, goals, objectives, strategies, and external and internal environment of the organisation. This would further be distilled to understanding the business issue in the context of the DM project and provide a solution that is consistent with the organisation business goals and objectives. This enables the miner to determine the best way to execute the project so that it will empower and facilitate the achievement of the business's targets. Failing to understand the organization's needs before implementing the project may cause its results to be of no use to the organization.

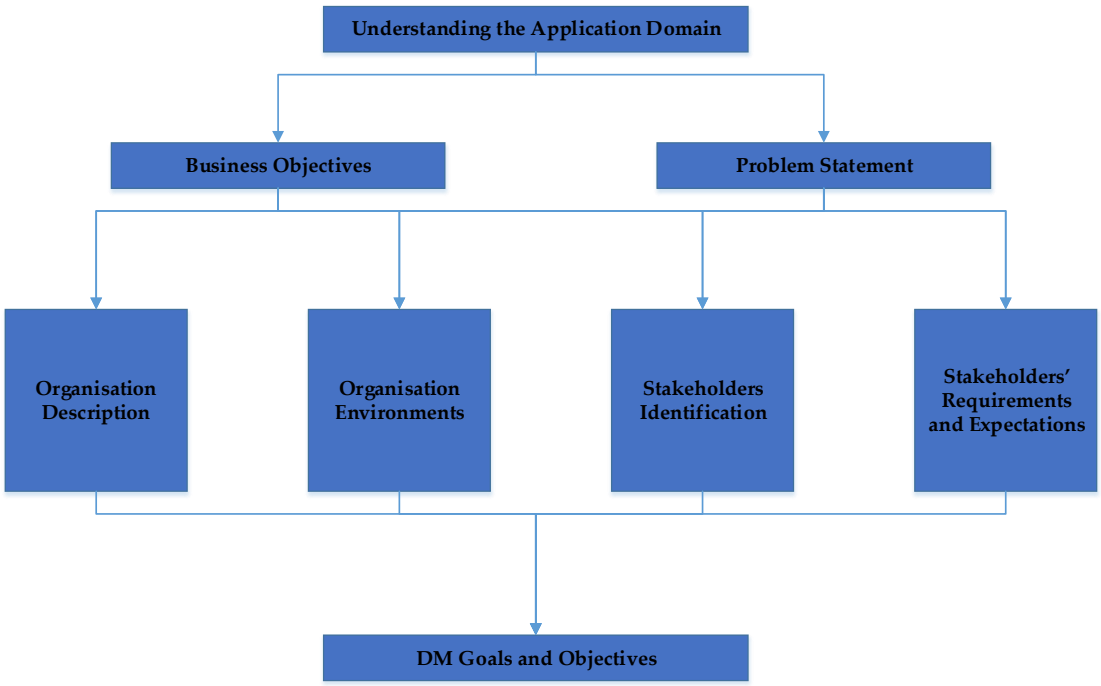


Figure 11: Organisational Analysis DM Methodology

4.2.1 Business Objectives

The business objectives otherwise called the model goals is the process of defining a relevant and realistic definition of the problem that we ensure it add value to the business. The two main objective for any businesses can be broadly categorised as:

1. Uncovering the trends and patterns
2. Proffering answers to the "why"

The business objective would ensure the organisation is able to relate the strategic objectives of a business to the use of the data mining.

4.2.2 Problem Statement

Problem identification is the most important stage in the entire process. There is need to critically assess the business situation quantitatively and qualitatively and to determine the importance given the overall business strategy. At this stage, we can best determine the overall relevance of the project to business, the current data environment and it's implication in resolving the business problems.

DM provides solution to a lot of problems that cannot be solved with traditional approaches; hence we must face these projects with a mindset free of bias. To be able to get this done, there is need to identify the problem we want to solve and the business benefits we want to obtain. This would form the basis of the project objectives, requirements, and preliminary implementation plan.

4.2.3 Organization Description

The purpose of DM is to provide is both descriptive and predictive in nature, the effectiveness of DM will depend on its understanding of the organisation This requires a broad and detailed description of the organisation. The company should assess its current operations, goals, practices, processes, and data, as well as any gaps, and fill them within the framework. For instance, the organisation should be able to state her purpose, state the business problems, determine if the DM project is for data analysis or predictive analysis, state if

she is data rich or data poor. This component of the framework must be customized to the needs of the organization. The organisation description must state both the problem and business objectives.

4.2.4 Organization Environments

When embarking on any DM project, the organization need to understand the external and internal environment it operates on.

The external factors include:

1. The social, cultural, political, legal, regulatory, financial, technological, economic and environmental factors, whether international, national, regional or local.
2. Key drivers and trends affecting the objectives of the organization.
3. External stakeholders' relationships, perceptions, values, needs and expectations.
4. Contractual relationships and commitments.

The internal factors include:

1. Vision, mission, and values.
2. Governance, organizational structure, roles, and responsibilities.
3. Strategy, objectives, and policies.
4. The organization's culture.

5. Standards, and operating procedures adopted by the organization.
6. Capabilities, understood in terms of resources and knowledge (e.g. capital, time, people, intellectual property, processes, systems and technologies).
7. Data and information systems
8. Contractual relationships and commitments.
9. Interdependencies and interconnections.

4.2.5 Stakeholders Identification

To successfully implement a DM project, it is very important to identify all the stakeholders of the project. Stakeholders include all the major owners, users, analysts, designers, and developers of an information system, as well as the other people who would be impacted by the project. Proper identification of the stakeholders will allow miner to completely recognize the critical elements of the project, together with its true intentions, expected results and its usage.

4.2.6 Stakeholders' Requirements and Expectations

To defining the goals and objectives of the DM project, it is proper to define the requirements and expectations of the stakeholders. DM is an information project as such there is need to get buy-in from all stakeholders; this would ensure their direct involvement and expectations are managed without compromising the DM goals and objectives.

4.2.7 DM Goals and Objectives

DM goals and objectives can be broadly classified as follows:

- **Predictive:** Determine how certain attributes will behave in the future.
- **Identification:** Identify patterns in data.
- **Classification:** Partition data into classes.
- **Optimization:** Optimize the use of limited resources such as time, space, money or materials.

The business objectives and problems statement and must be cascaded to a DM goals and objectives.

4.3 Integrated Project and Change Management

Project management ensures the project is designed, developed and delivered successfully while Change management ensures that the project is embraced, adopted and used by the targets or recipients of the change. Both must be managed in an integrated manner such that the DM outcome delivers the desired outcomes to the business. It must be dome from both the technical side and the people side.

To ensure value is realized in the shortest possible time, the DM project must make provision for constant feedback and change management which must be incorporated into every part of the project. DM is iterative in nature and is best

managed using Agile Methodologies which focuses on delivering maximum value against business priorities.

Change must be managed in the DM project otherwise the project would fall apart. Most DM project does not have the right data at the onset, thus there is a need to operationalize and prepare for change in DM project.

The integration creates value for the DM project by ensuring there are shared objectives of improving the organisational performance by successfully delivers the intended results and outcomes, ensure proactive steps to identify and mitigate project risks in a more proactive manner while addressing anticipated obstacles and resistance and build commitment and buy-in for the change. In addition, it allows for sequencing and alignment DM to produce the right outcomes for the project while improving the flow of information.

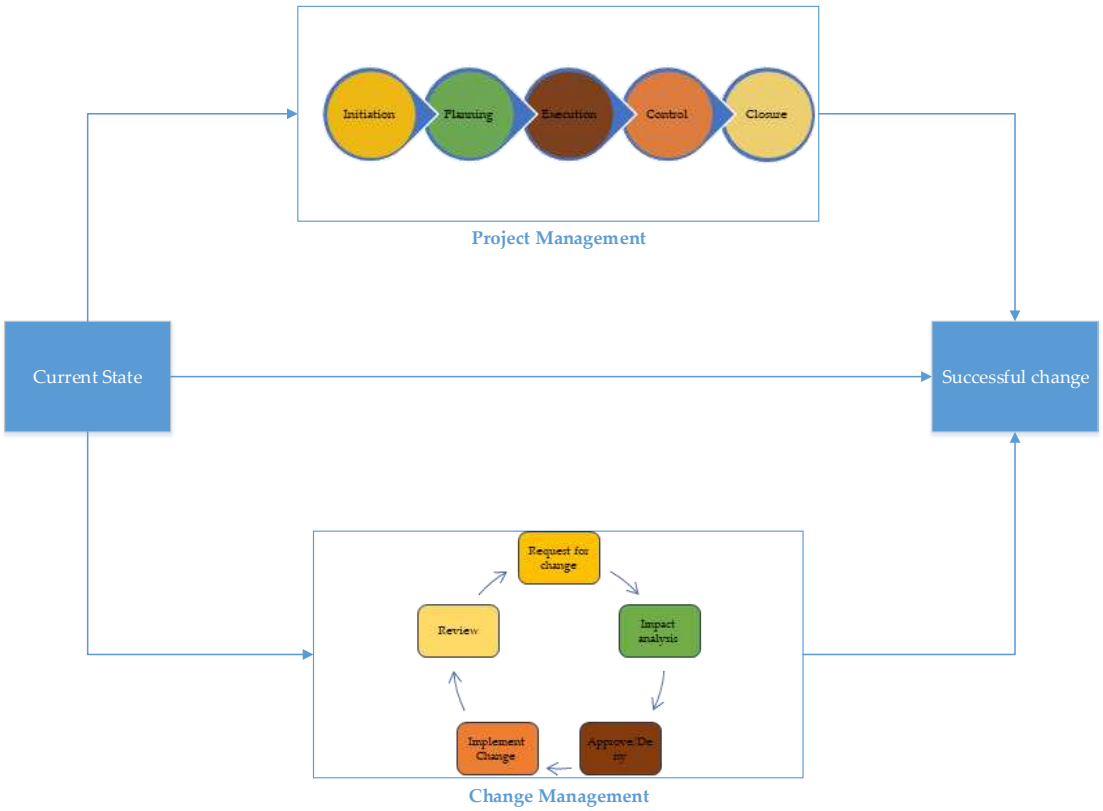


Figure 12: Integrated Project and Change Management

4.4 Data Mining

Data mining tasks involves a lot of process. All these collectively is aimed at ensuring the result or business objectives of the DM project is achieved.

4.4.1 Data Prospecting

DM projects assume that required information is already available; unfortunately, that is not always the case. A critical success factor is to identify the source of information for the DM project. In addition, it is important to consider some important aspect of the information such as:

- Data owners
- Data format

- Data retrieval cost
- Data size
- Data Privacy

GDPR is the toughest privacy and security law in the world Proton Technologies AG (2020) as such privacy and security are special issues that must be managed with extreme care. Regulations can confine the use of certain data or require special authorization to use it.

A good data documentation strategy must be in place to determine the type and format of data, the units of measurement to be used, the ranges of values and the primary keys of such data.

4.4.2 Data understanding

The miner needs to recognize and understand what has been gathered. The data needs to be understood, the relationships and patterns need to be uncovered so that other processes can commence. To perform a good DM analysis, the information available must be good. Errors, missing values, and other noise factors should be avoided. Data sets can be cleaned in data warehouses after it has been collected, but this process also implies the loss of information, which can help create better rules and models.

4.4.3 Data preparation

The data must be transformed in such a way as to correct inaccuracies, remove irregularities, eliminate duplicated data, detect and correct missing values, and check for any possible inconsistencies, before the analysis can began.

Good validation and consistency checks are best done at the source. This is not always easy as such many techniques would be applied on data to clean and prepare it for data mining analysis.

In the case of missing values, the approaches that are usually considered are the following Han, J., & Kamber, M., (2001):

- Ignore records.
- Fill in values manually.
- Create a special value or category.
- Use the mean value of the distribution.
- Use the mean value of the same class.
- Use the most probable value.

There is a lot of trade-off and bias that may result from using any of the approach above, all of them alter the record in one way or another, and further increases the degree of contamination already existing. If not handled with extreme caution based on the understanding of the data can result in building a worn model.

In order to eliminate noise in data, several other methods can be employed Han, J., & Kamber, M., (2001):

- Binning
- Clustering
- Combine inspection
- Regression

Binning analysis separates data values into buckets or bins and "smooths" them out by replacing the data in each bin with the group's mean, median, or closest border, as appropriate. Clustering can also be used to reduce noise by detecting extreme values among groups with comparable values. The combined inspection method requires the interaction between computer and human inspection. In this method, computers anticipate values that are compared to a set of thresholds; when values surpass the threshold, they are presented to a human user, who then determines whether the value may be accepted or not.

However, the regression method fits data to a certain function. Regression can be linear, multiple, or multidimensional depending on the number of variables used in the analysis. Although noise detection is an important element for accurate data mining analysis, it must be implemented with caution. In applications like process monitoring and quality control, noise reduction can

potentially erase crucial patterns and relationships among data items that must be discovered.

Other data transformations that can be useful for data mining analysis are aggregation, generalization, normalization and attribute constructions. With aggregation and generalization, data can be analyzed at different summarized and hierarchical levels.

4.4.4 Model Development

Model creation is an iterative process in which data mining tools evaluate data and build rules, patterns, and correlations. Human experts are required to identify, choose, and decide which are the most important rules and significant models based on the different algorithms used.

The algorithm must be trained, users need to understand how to manage the tools and techniques used. The miner must understand the process, task, material and conditions involved in the data to discover a truly insightful information.

In model development, data is explored to identify the most relevant fields. The data available is then divided randomly in subsets, at least one set for training and at least one set for validation. Models are created using the rules and relationships discovered according to the DM task and techniques selected for the project. Finally, the predictions of the models must be tested against the

validation sets. If the predictions are sufficiently good, the models can be implemented; otherwise, more models are explored in an iterative process.

4.4.5 Model Validation

Model validation phase is used to determine if the models created by the DM tools can correctly predict the behavior of the variables represented by the data. Thresholds can be applied to this activity based on the individual demands and conditions of each project. Cross-validation and bootstrapping are two validation techniques that can be used to estimate the errors of the models. Against ensure that the models are legitimate, the error values can be compared to thresholds.

It is not guaranteed that a model will always successfully predict the values of the variables represented by similar or fresh data if it successfully predicts the values in both the training and validation sets. For this reason, threshold values should be periodically revised according to the current requirements of the organization. Furthermore, additional testing with new data is required if the intention of the project is to predict the behavior of a real system.

4.4.6 Model Implementation

Once a model is validated, it can be implemented according to the business goals and objectives initially established for the DM project. Model implementation is an important phase and also requires analyzing and interpreting the results generated by the models. Not all data mining projects

require the implementation of a specific model. The information obtained during the process, as well as any rules, relationships, or patterns uncovered, can be utilized to solve specific problems, make suggestions, make judgments, or determine the need for additional research.

If the models are implemented with other applications, implementation itself can be considered as a part of a system analysis and design process, and it would require additional testing. Models can be used to classified specific records, assign probabilities, or generate special orders or reports.

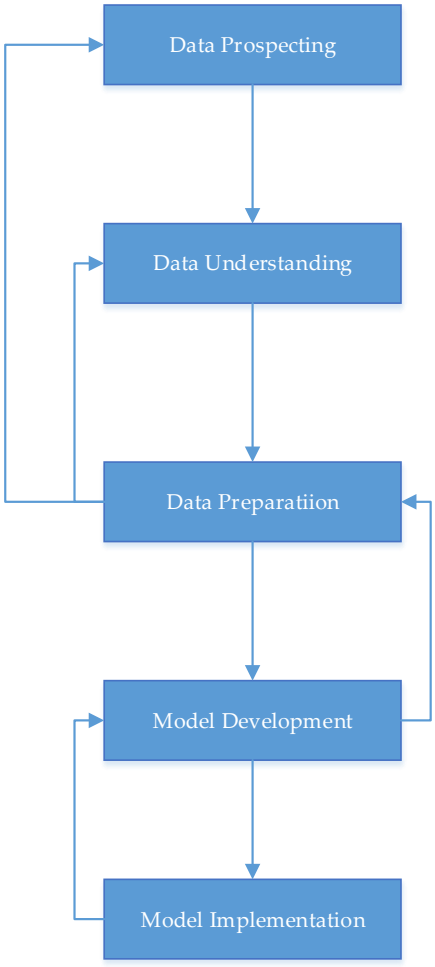


Figure 13: Data Mining Stages

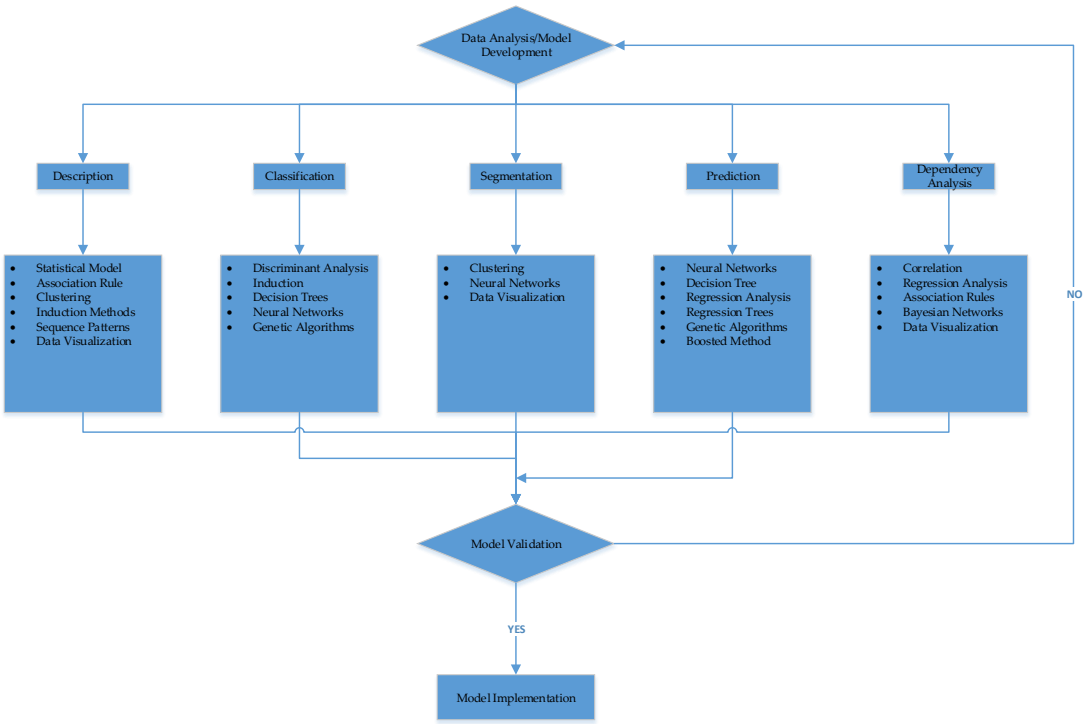


Figure 14: Model Development

4.5 Interpretation

Data interpretation refers to the application of procedures for analyzing data in order to make an informed choice. The interpretation of data gives the information studied a meaning and determines its significance and ramifications.

Data interpretation is the end product of DM project and provides mean for all of the efforts and that is why it must be done properly. Data is likely to come from a variety of sources and enter the analysis process in a haphazard order. DM project are very subjective since the same data can be analysed and use for varying business needs.

Data miner must discern the differences between correlation, causation and coincidences when interpreting data in addition to removing all bias, while putting into consideration all factors that led to the result. as well as a variety of other biases – but he must also analyze all of the circumstances that may have contributed to the outcome. One can employ a variety of data interpretation techniques. People can make sense of numerical data that has been collected, processed, and presented by interpreting it. The organisation must put in place a baseline method for interpreting data will serve as a solid structure and consistent foundation for all interpretation processes. Data are interpreted in the two broadest and most common categories namely:

1. Quantitative analysis
2. Qualitative analysis

4.5.1 Qualitative Data Interpretation

Qualitative data analysis is the process of categorizing the data. Qualitative analysis describe data using descriptive context (i.e., text) rather than numbers or patterns to describe data. Narrative data is gathered by employing a wide variety of techniques such as:

- Observations
- Documents
- Interviews

4.5.2 Quantitative Data Interpretation

To assess numerical data, the quantitative data interpretation approach is applied. Due to its natural existence as a number, miners do not need to employ the coding technique on quantitative data before it is analyzed. Statistical modeling approaches such as standard deviation, mean, and median are used in the analysis of quantitative data. The following are some of them:

- Mean
- Standard deviation
- Frequency distribution
- Regression analysis
- Cohort analysis
- Predictive and prescriptive analysis

4.6 Presentation and Visualization

Data interpretation is best when there is a visualization of the dataset. Visualization is a natural way to explain the dataset and ensure anyone understand the data with a visually appealing summary of the data.

4.7 Integration, Monitoring and Review

For DM projects and activities to be effective and the organisation to benefit from the investment on the long run, it must be integrated into the organisation,

part of which is to ensure there is proper monitoring and review of DM efforts, the result, and the competitive gains of such initiatives.

Monitoring and review would provide the required assurance and improve the quality and effectiveness of process, implementation, and outcomes. Ongoing monitoring and periodic review of the DM process and its outcomes should be a planned part of the DM project, with clearly defined responsibilities. Some of the processes are maintenance operation on the equipment, data and information residing in data warehouses must be kept up to date, protected from unauthorized access and that backups are done regularly.

The results of monitoring and review should be incorporated throughout the organization's performance management, measurement, and reporting activities.

4.8 Conclusion

By using a systems analysis approach, this chapter presented a proposed methodology for using data mining in solving problems related to industrial engineering. Analyze the organization, arrange the task, build a data model, implement the model, and provide ongoing support are the five primary steps of the suggested technique. Each of these phases has been detailed in detail, and they cover the essential procedures that any data mining project in industrial engineering must follow from conception through completion and support. The proposed technique provides a strong foundation for industrial

engineers to use data mining in a consistent and repeatable manner, allowing them to evaluate data mining projects, replicate results, and pinpoint where problems in their data mining initiatives occurred.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

The key to long term profitability of an insurer is to strategically address risk pooling, eliminate fraud and a data driven standards of CRM in such a way as to provide stable, high-value insurance policies that provide risk protection in an effective manner. Many of the strategy in use today fail to address these issues because they are not data driven.

Using the hidden patterns and relationship discovered by applying DM methodology, tools and techniques, insurer would discover useful information that can lead to better understanding of their business, systems, and processes. This information can then be used to draft new policies, develop better standard operating procedures, design new processes, new products and embed DM into their processes so they can always make informed decision that would lead to better performance and resource utilization. This study's methods may be useful in these endeavors.

5.1 Application in Life Insurance

DM has its root in serving as expert system to uncover patterns and relationships for problems that cannot rely on assumption. The scope has expanded, and it is now used to achieve other goals. DM methodology is an important perspective that must be considered when applying DM to

insurance. Customer data is now a critical asset for any insurer. The traditional approach of DM has been from the statistical point of view. We require a system-oriented framework, that has the fundamental components of an information system, managed as a project, integrated into the overall corporate processes, one that is monitored and reviewed frequently based on the environmental variables, and provide a implementation-neutral approach to DM.

Most traditional approach to DM generally does not consider the roles of the organization and the stakeholders during the project, it treats data mining as a separate element of the organisational systems and lay more emphasis on a specific software to accomplish the task. It does not consider data mining to be an important part of the organizational system. DM methodology should not be software-dependent and must ensure that there is sufficient information flow within the organisation.

The DM method should be flexible enough to embrace any tools and techniques, but only after the business requirements of the organization have been well documented and understood. Selecting tools and techniques very early create a risk of rendering the data mining effort useless as the model may not be truly representational of the business goals.

DM techniques must follow the organisational goal and data available. Decision around data availability to carry out the project must be taken in the

beginning of the project. Every organisation can determine if they are data rich or data poor and would thus make effort to gather all data that would ensure the model fit into the organisation's requirements. This methodology makes no assumption in that regard and readily allow for insurer to analyses existing data stored in data warehouse or data mart while pro-actively source for data to augment existing data stores or creating new ones to make needed data accessible.

In addition, there is need to ensure that DM is not a one-off approach, the change in the marketplace is high as such insurer should be able to cope with the dynamics of the market while putting in place ways of ensuring the organisation become data rich, these data must be complete and of high quality so that subsequent DM effort can build on the foundation of earlier projects. In all of these, DM task must use an agile approach since it allows for frequent feedbacks, speed to market and managing changes and further refinement during the project this would further ensure insurer have capacity to be agile which is seriously lacking in the insurance industry.

Documentation is an essential part of DM projects because the models created with rules extracted using data mining tools can be very complex. As a result, a good documentation effort is required during the entire project is required. This would further assist miners to keep track of all changes performed and to make all the modifications needed for the corresponding models. This

documentation not only serves as a description of the data repositories as well as the data mining activities and their outcomes, but it also acts as a baseline for future improvements.

5.2 Application Concern

DM projects using data mining tools and techniques are not always easy to implement and may require considerable amounts of time and resources. Insurer needs to give attention to planning and preparation for the project so that there can be better chance of success and usefulness of the result.

Not all data mining projects produce useful results as such all assumptions must be verified, models must be subjected to various testing such that it would truly represent the behavior of the organisation's problems.

Applying DM in insurance can result a wide variety of benefits for organizations. They can, for example, completely use data on customers' buying patterns and behavior in order to obtain a better knowledge of client motives in order to prevent fraud, anticipate resource need, increase acquisition, and reduce customer churn.

DM is a tool for production useful information, as such the result may vary depending on how the DM is applied and the skill of the miner.

5.3 Application Issues

As previously stated, data mining applications in manufacturing and other fields of industrial engineering necessitate a thorough knowledge of the

processes and systems involved. As a result, data mining analysts must not only have good awareness of and familiarity with all the processes and variables of the study, but also, they must be certain that all of the data sources available are valid, correspond to the specifications and requirements, and are suitable for analysis.

Consequently, the training and preparation required for these types of studies may be extensive--not only for the selection of the corresponding tasks and techniques, but also for the application of the necessary tools. Many data mining tools and solutions are complex and difficult to use; they offer a large range of methods, jobs, and models to choose from. Therefore the training required for data mining projects must also include training in data mining software packages, which may take 80 hours or longer.

Additionally, not all companies and organizations are prepared to conduct data mining studies. Companies may not have considered the possibility of using data mining in their records and historical information. As a result, when the opportunity to use data mining techniques arises, it's possible that not all of the necessary data is available. It's also possible that the data is dispersed over a variety of databases, locations, and formats. These conditions significantly degrade data mining performance, require the application of special data cleaning and data preparation techniques, and represent a considerable new investment of time and other resources. The ideal solution to

this problem is to construct databases, data marts, data repositories, and data warehouses in such a way that data mining may be applied successfully to meet the needs and expectations of all stakeholders.

Another consideration is that not all available data is useful, and historical data may not generate good rules or prediction models. Some processes in manufacturing are so dynamic and flexible that the information available may not correspond with existing conditions or current products and processes. Additionally, in some cases information may no longer be relevant, and analysts may be unaware of the data's currency. Wrong assumptions may be made, thereby, generating erroneous or inaccurate models that lead to inappropriate recommendations.

As a result, extreme caution is advised while picking data sources for investigations. Data mining analysts in industrial engineering projects are required to have a good understanding of the process and the selected tools and techniques. They also need to know where the data came from before starting the data mining process.

Only when these requirements are met will the rules, models, interactions, and relationships found in the studies have a greater probability of representing the real systems. Analysts then can confidently appreciate what the information implies.

Finally, data mining initiatives might be extremely costly. Costs do not include only the value of the software packages (which ranges at present from $100 to $250,000 and can be segregated in many different modules), but also other related expenses, such as additional annual licensing fees, software, hardware, installation, training, technical support, maintenance, consulting, and outsourcing. These costs include up-front, one-time costs as well as on-going costs. As a result, companies must examine the data mining project's economic feasibility throughout its useful life.

5.4 Future Work

There are many areas for the application of DM in insurance, although many of them have been listed in this research, many others can also be found. The methodology proposed in this research provides a structural basis from which additional studies can be developed.

This research presented a conceptual model to be applied in life insurance DM project, it would surely be applicable to a variety of other industry and DM project. The next phase in this research is to put this conceptual model to the test and improve it.

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